

48. $\Delta\Sigma$ Interface (DSMIF)

48.1 Overview

The $\Delta\Sigma$ interfaces (DSMIF) can be connected with up to three external $\Delta\Sigma$ modulators. It is capable of filtering delta-sigma modulated 1-bit digital input data and converting it into 16-bit digital data.

Table 48.1 lists the specifications of DSMIF and Table 48.2 lists the functions of DSMIF. Figure 48.1 shows a block diagram of DSMIF.

Table 48.1 DSMIF specifications

Item	Description
Number of units	2 units
Number of channels	3 channels per unit for a total of 6 channels IP
Core clock (CCLK)*1	Selectable from PCLKA, DSMIFCLK or GPTCLK
Functions	- Capable of filtering 1-bit digital input data DSMDATAN (n = 0 to 2) and converting them into 16-bit digital values. - The sinc filters are selectable as first-, second-, or third-order. - DSMCLKn can be output at up to 25 MHz in master operation and input at up to 25 MHz in slave operation.*1 - Current data can be captured with the same timing by capture trigger. - Error detect functions can be used.
Interrupt sources	12 kinds of interrupt sources in 1 unit 1. Current data register update interrupt per channel (Pulse) 2. Current capture data register A update interrupt per channel (Pulse) 3. Current capture data register B update interrupt per channel (Pulse) 4. Short-circuit detection error interrupt per channel (Level) 5. Overcurrent lower limit detection 0 error interrupt per channel (Level) 6. Overcurrent upper limit detection 0 error interrupt per channel (Level) 7. Overcurrent lower limit detection 1 error interrupt per channel (Level) 8. Overcurrent upper limit detection 1 error interrupt per channel (Level) 9. Overcurrent lower limit detection 2 error interrupt per channel (Level) 10. Overcurrent upper limit detection 2 error interrupt per channel (Level) . 11. Sum error lower limit detection interrupt (Level) 12. Sum error upper limit detection interrupt (Level) .
Event link function (input)	Six inputs for each channel Current data can be transferred to other registers in response to event signals used as triggers.
Notification function (output)	4 notifications are output from each DSMIF's channel The result of comparison with the upper and lower limits of overcurrent detection is output as an event notification.
PWM synchronization function (input)	Three inputs for each channel The PWM synchronization function is realized by initializing the decimation counter for the differential stage of the SINC filter with the trigger output from GPT.
Low-power consumption function	This module can be placed in the module stop state.
TrustZone filter	Security and Privilege attribution can be set for each unit.

Note 1. Clock frequency ratio should be $CCLK/5 \leq PCLKA$, $CCLK/6 \geq DSMCLKn$.

Table 48.2 DSMIF functions (1 of 2)

Parameter			Unit 0 (DSMIF0)	Unit 1 (DSMIF1)
I/O pin	Clock	Clock I/O pin CHn	DSM0CLKn	DSM1CLKn
	Data	DATA input pin CHn	DSM0DATAn	DSM1DATAn

Table 48.2 DSMIF functions (2 of 2)

Parameter			Unit 0 (DSMIF0)	Unit 1 (DSMIF1)
Interrupt source	Normal	Current data register update interrupt CHn	DSMIF0_CDUPDn	DSMIF1_CDUPDn
		Current capture data register A update interrupt CHn	DSMIF0_CDAUPDn	DSMIF1_CDAUPDn
		Current capture data register B update interrupt CHn	DSMIF0_CDBUPDn	DSMIF1_CDBUPDn
		Current data register update common interrupt	DSMIF0_CDUPD_COM	DSMIF1_CDUPD_COM
		Current capture data register A update common interrupt	DSMIF0_CDACUPD_COM	DSMIF1_CDACUPD_COM
		Current capture data register B update common interrupt	DSMIF0_CDBCUPD_COM	DSMIF1_CDBCUPD_COM
	Error	Overcurrent sum error lower limit detection interrupt	DSMIF0_ERR	DSMIF1_ERR
		Overcurrent sum error upper limit detection interrupt		
		Overcurrent lower limit detection 0 interrupt CHn		
		Overcurrent upper limit detection 0 interrupt CHn		
		Overcurrent lower limit detection 1 interrupt CHn		
		Overcurrent upper limit detection 1 interrupt CHn		
		Overcurrent lower limit detection 2 interrupt CHn		
		Overcurrent upper limit detection 2 interrupt CHn		
		Short circuit detection interrupt CHn		
Trigger input ELC destination	Capture	Current data capture trigger 0	DSMIF0 CAP_TRG0	DSMIF1 CAP_TRG0
		Current data capture trigger 1	DSMIF0 CAP_TRG1	DSMIF1 CAP_TRG1
		Current data capture trigger 2	DSMIF0 CAP_TRG2	DSMIF1 CAP_TRG2
		Current data capture trigger 3	DSMIF0 CAP_TRG3	DSMIF1 CAP_TRG3
		Current data capture trigger 4	DSMIF0 CAP_TRG4	DSMIF1 CAP_TRG4
		Current data capture trigger 5	DSMIF0 CAP_TRG5	DSMIF1 CAP_TRG5
	Counter Reset	Decimation dividing counter initialization trigger 0	DSMIF0 CDCNT_INT_TRG0	DSMIF1 CDCNT_INT_TRG0
		Decimation dividing counter initialization trigger 1	DSMIF0 CDCNT_INT_TRG1	DSMIF1 CDCNT_INT_TRG1
		Decimation dividing counter initialization trigger 2	DSMIF0 CDCNT_INT_TRG2	DSMIF1 CDCNT_INT_TRG2
POEG output disable request	Overcurrent detection window notification 0 CHn		DSMIF0_OVC_WND0_NTCn	DSMIF1_OVC_WND0_NTCn
	Overcurrent detection window notification 1 CHn		DSMIF0_OVC_WND1_NTCn	DSMIF1_OVC_WND1_NTCn
	Overcurrent detection window notification 2 CHn		DSMIF0_OVC_WND2_NTCn	DSMIF1_OVC_WND2_NTCn
	Overcurrent detection window notification 3 CHn *1		DSMIF0_OVC_WND3_NTCn	DSMIF1_OVC_WND3_NTCn

Note: n = 0 to 2

Note 1. Bound unit error signal connected to POE3 and POEG as error 0

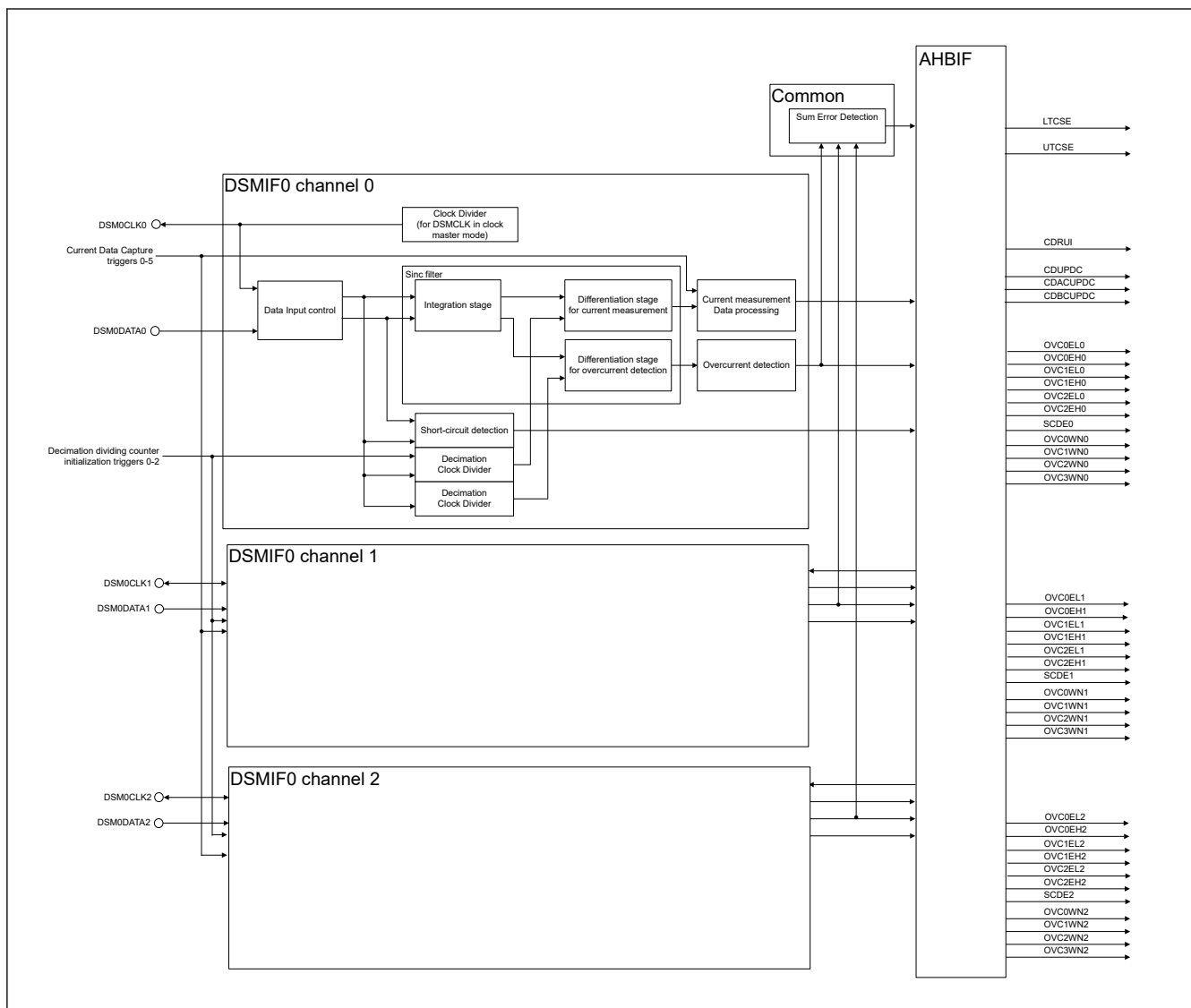


Figure 48.1 Block diagram of DSMIF0

48.2 Register Descriptions

48.2.1 DSMIF Common Control Registers

48.2.1.1 DSCCSCR : Core Clock Selection Control Register

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m(m = 0, 1)
 DSMIFm_NS = 0x5032_A000 + 0x0800 × m(m = 0, 1)

Offset address: 0x000

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CLKSEL[1:0]
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	CLKSEL[1:0]	Selection of core clock (CCLK) 0 0: PCLKA 0 1: DSMIFCLK 1 0: GPTCLK 1 1: Setting prohibited	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.1.2 DSSEICR : Overcurrent Sum Error Detect Interrupt Control Register

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x080

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ISEH	ISEL
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ISEL	Overcurrent sum error lower limit detection interrupt enable bit 0: Do not output overcurrent sum error lower limit detection interrupt 1: Overcurrent sum error lower limit detection interrupt can be output	R/W
1	ISEH	Overcurrent sum error upper limit detection interrupt enable bit 0: Do not output overcurrent sum error upper limit detection interrupt 1: Overcurrent sum error lower upper detection interrupt can be output	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.1.3 DSSECSR : Overcurrent Sum Error Detect Channel Setting Register

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x084

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	SEDM[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	SEDM[2:0]	Overcurrent sum error detect mode setting bit 0 0 0: Detects error of sum value of overcurrent data of CH0, CH1, CH2 0 0 1: Detects error of sum value of overcurrent data of CH0, CH1 0 1 0: Detects error of overcurrent data of CH0 0 1 1: Detects error of overcurrent data of CH1 1 0 0: Detects error of overcurrent data of CH2 Others: Setting prohibited (detects error of sum value of overcurrent data of CH0, CH1, CH2)	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.1.4 DSSELTR : Overcurrent Sum Error Detect Low Threshold Register

Base address: $\text{DSMIF}_m = 0x4032\ A000 + 0x0800 \times m$ ($m = 0, 1$)

Offset address: 0x088

[illegible]

Bit	Symbol	Function	R/W
17:0	SCMPTBL[17:0]	Overcurrent sum error detect lower limit These bits set the lower limit of the active current value for overcurrent sum error detection. Set these bits to the lower limit of the sum value of the overcurrent data for each channel.	R/W
31:18	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.1.5 DSSEHTR : Overcurrent Sum Error Detect High Threshold Register

Base address: $DSMIF_m = 0x4032_A000 + 0x0800 \times m$ ($m = 0, 1$)

Offset address: 0x08C

[illegible]

Bit	Symbol	Function	R/W
17:0	SCMPTBH[17:0]	Overcurrent sum error detect upper limit These bits set the upper limit of the active current value for overcurrent sum error detection. Set these bits to the upper limit of the sum value of the overcurrent data for each channel.	R/W
31:18	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.1.6 DSSECR : Overcurrent Sum Error Detect Control Register

Base address: $DSMIF_m = 0x4032\ A000 + 0x0800 \times m$ ($m = 0, 1$)

Offset address: 0x090

[illegible]

Bit	Symbol	Function	R/W
0	SEEL	Overcurrent sum error lower limit detection enable bit 0: The lower limit exceeded detection function of overcurrent sum error detection is invalid 1: The lower limit exceeded detection function of overcurrent sum error detection is valid	R/W
1	SEEH	Overcurrent sum error upper limit detection enable bit 0: The upper limit exceeded detection function of overcurrent sum error detection is invalid 1: The upper limit exceeded detection function of overcurrent sum error detection is valid	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.1.7 DSSECDRn : Overcurrent Sum Error Detect Capture Data Register n (n = 0 to 2)

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x0A0 + 0x04 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SECDR[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	SECDR[15:0]	Overcurrent sum error detect capture data n Captures the overcurrent data of the target channel when the upper limit error or the lower limit error of the over sum current error detection occurs. See Table 48.9 for details of the data to be captured.	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.1.8 DSCMSR : Common Mode Setting Register

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x0C0

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	CISM[1:0]	DFS	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DFS	Data Format Select 0: Left justified 1: Right justified	R/W

Bit	Symbol	Function	R/W
2:1	CISM[1:0]	Common Interrupt Synchronous channel Mode 0 0: Common interrupts are not used 0 1: Common interrupts for ch0, ch1, and ch2 (ch3, ch4, and ch5) can be used 1 0: Common interrupts for ch0 and ch1 (ch3 and ch4) can be used 1 1: Setting prohibited	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.1.9 DSCICR : Common Interrupt Control Register

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x0C4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	IBUCE	IAUCE	IUCE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	IUCE	Current data register update channel common interrupt enable 0: Do not output capture current data register update channel common interrupt 1: Capture current data register update channel common interrupt can be output	R/W
1	IAUCE	Capture current data register A update channel common interrupt enable 0: Do not output capture current data register A update channel common interrupt 1: Capture current data register A update channel common interrupt can be output	R/W
2	IBUCE	Capture current data register B update channel common interrupt enable 0: Do not output capture current data register B update channel common interrupt 1: Capture current data register B update channel common interrupt can be output	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.1.10 DSCSTRTR : Channel Software Start Trigger Register

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x200

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	STRT RG2	STRT RG1	STRT RG0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	STRTRG0	Channel 0 start trigger 0: Do nothing 1: Start channel	W

Bit	Symbol	Function	R/W
1	STRTRG1	Channel 1 start trigger 0: Do nothing 1: Start channel	W
2	STRTRG2	Channel 2 start trigger 0: Do nothing 1: Start channel	W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.1.11 DSCSTPTR : Channel Software Stop Trigger Register

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x204

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	STPT RG2	STPT RG1	STPT RG0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	STPTRG0	Channel 0 stop trigger 0: Do nothing 1: Stop channel	W
1	STPTRG1	Channel 1 stop trigger 0: Do nothing 1: Stop channel	W
2	STPTRG2	Channel 2 stop trigger 0: Do nothing 1: Stop channel	W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.1.12 DSCESR : Channel Error Status Register

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x220

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SUME RRH	SUME RRL
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	SCF2	SCF1	SCF0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SCF0	Channel 0 short circuit detection flag This bit becomes 1 when 1 is continuously input to the DSMDATA0 pin, exceeding the number of times set in DSSCTSRCH0.SCNTH, or when 0 is continuously input to the DSMDATA0 pin, exceeding the number of times set in DSSCTSRCH0.SCNTH. This bit becomes 0 when SCF0 = 1 and writing 1 to the channel 0 short circuit detection flag clear (DSCESCR.CLRSCF0 / DSCSCRCH0.CLRSCF). 0: Indicates that the DSMDATA0 pin short circuit is not detected 1: Indicates that a short circuit on the DSMDATA0 pin has been detected	R
1	SCF1	Channel 1 short circuit detection flag This bit becomes 1 when 1 is continuously input to the DSMDATA1 pin, exceeding the number of times set in DSSCTSRCH1.SCNTH, or when 0 is continuously input to the DSMDATA1 pin, exceeding the number of times set in DSSCTSRCH1.SCNTH. This bit becomes 0 when SCF1 = 1 and writing 1 to the channel 1 short circuit detection flag clear (DSCESCR.CLRSCF1 / DSCSCRCH1.CLRSCF). 0: Indicates that the DSMDATA1 pin short circuit is not detected 1: Indicates that a short circuit on the DSMDATA1 pin has been detected	R
2	SCF2	Channel 2 short circuit detection flag This bit becomes 1 when 1 is continuously input to the DSMDATA2 pin, exceeding the number of times set in DSSCTSRCH2.SCNTH, or when 0 is continuously input to the DSMDATA2 pin, exceeding the number of times set in DSSCTSRCH2.SCNTH. This bit becomes 0 when SCF2 = 1 and writing 1 to the channel 2 short circuit detection flag clear (DSCESCR.CLRSCF2 / DSCSCRCH2.CLRSCF). 0: Indicates that the DSMDATA2 pin short circuit is not detected 1: Indicates that a short circuit on the DSMDATA2 pin has been detected	R
15:3	—	These bits are read as 0.	R
16	SUMERRL	Overcurrent sum error lower limit detection flag This bit becomes 1 when the sum value of the overcurrent data is updated and this sum value is smaller than the lower limit (DSSELTR.SCMPTBL). This bit becomes 0 when SUMERRL = 1 and writing 1 to the overcurrent sum error lower limit detection flag clear (DSCESCR.CLRSUMERRL). 0: Indicates that the sum value of overcurrent data is not less than the lower limit 1: Indicates that the sum value of overcurrent data is less than the lower limit	R
17	SUMERRH	Overcurrent sum error upper limit detection flag This bit becomes 1 when the sum value of the overcurrent data is updated and this sum value is larger than the upper limit (DSSEHTR.SCMPTBH). This bit becomes 0 when SUMERRH = 1 and writing 1 to the overcurrent sum error upper limit detection flag clear (DSCESCR.CLRSUMERRH). 0: Indicates that the sum value of overcurrent data is not more than the upper limit 1: Indicates that the sum value of overcurrent data is more than the upper limit	R
31:18	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

48.2.1.13 DSCOCESR : Channel Overcurrent Error Status Register

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x224

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	OC2F H2	OC2F H1	OC2F H0	—	OC2F L2	OC2F L1	OC2F L0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	OC1F H2	OC1F H1	OC1F H0	—	OC1F L2	OC1F L1	OC1F L0	—	OC0F H2	OC0F H1	OC0F H0	—	OC0F L2	OC0F L1	OC0F L0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	OC0FL0	Channel 0 overcurrent lower limit detection 0 flag This bit becomes 1 when the overcurrent data is updated and this data is smaller than the lower limit 0. This bit becomes 0 when OC0FL0 = 1 and writing 1 to the channel 0 overcurrent lower limit detection flag 0 clear (DSCOCESCR.CLROC0FL0 / DSCSCRCH0.CLROC0FL). 0: Indicates that the overcurrent data is not less than the lower limit 0 1: Indicates that the overcurrent data has been less than the lower limit 0	R
1	OC0FL1	Channel 1 overcurrent lower limit detection 0 flag This bit becomes 1 when the overcurrent data is updated and this data is smaller than the lower limit 0. This bit becomes 0 when OC0FL1 = 1 and writing 1 to the channel 1 overcurrent lower limit detection flag 0 clear (DSCOCESCR.CLROC0FL1 / DSCSCRCH1.CLROC0FL). 0: Indicates that the overcurrent data is not less than the lower limit 0 1: Indicates that the overcurrent data has been less than the lower limit 0	R
2	OC0FL2	Channel 2 overcurrent lower limit detection 0 flag This bit becomes 1 when the overcurrent data is updated and this data is smaller than the lower limit 0. This bit becomes 0 when OC0FL2 = 1 and writing 1 to the channel 2 overcurrent lower limit detection flag 0 clear (DSCOCESCR.CLROC0FL2 / DSCSCRCH2.CLROC0FL). 0: Indicates that the overcurrent data is not less than the lower limit 0 1: Indicates that the overcurrent data has been less than the lower limit 0	R
3	—	This bit is read as 0.	R
4	OC0FH0	Channel 0 overcurrent upper limit exceeded 0 flag This bit becomes 1 when the overcurrent data is updated and this data is larger than the upper limit 0. This bit becomes 0 when OC0FH0 = 1 and writing 1 to the channel 0 overcurrent upper limit exceeded flag 0 clear (DSCOCESCR.CLROC0FH0 / DSCSCRCH0.CLROC0FH). 0: Indicates that the overcurrent data doesn't exceed the upper limit 0 1: Indicates that the overcurrent data has exceeded the upper limit 0	R
5	OC0FH1	Channel 1 overcurrent upper limit exceeded 0 flag This bit becomes 1 when the overcurrent data is updated and this data is larger than the upper limit 0. This bit becomes 0 when OC0FH1 = 1 and writing 1 to the channel 1 overcurrent upper limit exceeded flag 0 clear (DSCOCESCR.CLROC0FH1 / DSCSCRCH1.CLROC0FH). 0: Indicates that the overcurrent data doesn't exceed the upper limit 0 1: Indicates that the overcurrent data has exceeded the upper limit 0	R
6	OC0FH2	Channel 2 overcurrent upper limit exceeded 0 flag This bit becomes 1 when the overcurrent data is updated and this data is larger than the upper limit 0. This bit becomes 0 when OC0FH2 = 1 and writing 1 to the channel 2 overcurrent upper limit exceeded flag 0 clear (DSCOCESCR.CLROC0FH2 / DSCSCRCH2.CLROC0FH). 0: Indicates that the overcurrent data doesn't exceed the upper limit 0 1: Indicates that the overcurrent data has exceeded the upper limit 0	R
7	—	This bit is read as 0.	R
8	OC1FL0	Channel 0 overcurrent lower limit detection 1 flag This bit becomes 1 when the overcurrent data is updated and this data is smaller than the lower limit 1. This bit becomes 0 when OC1FL0 = 1 and writing 1 to the channel 0 overcurrent lower limit detection flag 1 clear (DSCOCESCR.CLROC1FL0 / DSCSCRCH0.CLROC1FL). 0: Indicates that the overcurrent data is not less than the lower limit 1 1: Indicates that the overcurrent data has been less than the lower limit 1	R
9	OC1FL1	Channel 1 overcurrent lower limit detection 1 flag This bit becomes 1 when the overcurrent data is updated and this data is smaller than the lower limit 1. This bit becomes 0 when OC1FL1 = 1 and writing 1 to the channel 1 overcurrent lower limit detection flag 1 clear (DSCOCESCR.CLROC1FL1 / DSCSCRCH1.CLROC1FL). 0: Indicates that the overcurrent data is not less than the lower limit 1 1: Indicates that the overcurrent data has been less than the lower limit 1	R

Bit	Symbol	Function	R/W
10	OC1FL2	Channel 2 overcurrent lower limit detection 1 flag This bit becomes 1 when the overcurrent data is updated and this data is smaller than the lower limit 1. This bit becomes 0 when OC1FL2 = 1 and writing 1 to the channel 2 overcurrent lower limit detection flag 1 clear (DSCOCESCR.CLROC1FL2 / DSCSCRCH2.CLROC1FL). 0: Indicates that the overcurrent data is not less than the lower limit 1 1: Indicates that the overcurrent data has been less than the lower limit 1	R
11	—	This bit is read as 0.	R
12	OC1FH0	Channel 0 overcurrent upper limit exceeded 1 flag This bit becomes 1 when the overcurrent data is updated and this data is larger than the upper limit 1. This bit becomes 0 when OC1FH0 = 1 and writing 1 to the channel 0 overcurrent upper limit exceeded flag 1 clear (DSCOCESCR.CLROC1FH0 / DSCSCRCH0.CLROC1FH). 0: Indicates that the overcurrent data doesn't exceed the upper limit 1 1: Indicates that the overcurrent data has exceeded the upper limit 1	R
13	OC1FH1	Channel 1 overcurrent upper limit exceeded 1 flag This bit becomes 1 when the overcurrent data is updated and this data is larger than the upper limit 1. This bit becomes 0 when OC1FH1 = 1 and writing 1 to the channel 1 overcurrent upper limit exceeded flag 1 clear (DSCOCESCR.CLROC1FH1 / DSCSCRCH1.CLROC1FH). 0: Indicates that the overcurrent data doesn't exceed the upper limit 1 1: Indicates that the overcurrent data has exceeded the upper limit 1	R
14	OC1FH2	Channel 2 overcurrent upper limit exceeded 1 flag This bit becomes 1 when the overcurrent data is updated and this data is larger than the upper limit 1. This bit becomes 0 when OC1FH2 = 1 and writing 1 to the channel 2 overcurrent upper limit exceeded flag 1 clear (DSCOCESCR.CLROC1FH2 / DSCSCRCH2.CLROC1FH). 0: Indicates that the overcurrent data doesn't exceed the upper limit 1 1: Indicates that the overcurrent data has exceeded the upper limit 1	R
15	—	This bit is read as 0.	R
16	OC2FL0	Channel 0 overcurrent lower limit detection 2 flag This bit becomes 1 when the overcurrent data is updated and this data is smaller than the lower limit 2. This bit becomes 0 when OC2FL0 = 1 and writing 1 to the channel 0 overcurrent lower limit detection flag 2 clear (DSCOCESCR.CLROC2FL0 / DSCSCRCH0.CLROC2FL). 0: Indicates that the overcurrent data is not less than the lower limit 2 1: Indicates that the overcurrent data has been less than the lower limit 2	R
17	OC2FL1	Channel 1 overcurrent lower limit detection 2 flag This bit becomes 1 when the overcurrent data is updated and this data is smaller than the lower limit 2. This bit becomes 0 when OC2FL1 = 1 and writing 1 to the channel 1 overcurrent lower limit detection flag 2 clear (DSCOCESCR.CLROC2FL1 / DSCSCRCH1.CLROC2FL). 0: Indicates that the overcurrent data is not less than the lower limit 2 1: Indicates that the overcurrent data has been less than the lower limit 2	R
18	OC2FL2	Channel 2 overcurrent lower limit detection 2 flag This bit becomes 1 when the overcurrent data is updated and this data is smaller than the lower limit 2. This bit becomes 0 when OC2FL2 = 1 and writing 1 to the channel 2 overcurrent lower limit detection flag 2 clear (DSCOCESCR.CLROC2FL2 / DSCSCRCH2.CLROC2FL). 0: Indicates that the overcurrent data is not less than the lower limit 2 1: Indicates that the overcurrent data has been less than the lower limit 2	R
19	—	This bit is read as 0.	R
20	OC2FH0	Channel 0 overcurrent upper limit exceeded 2 flag This bit becomes 1 when the overcurrent data is updated and this data is larger than the upper limit 2. This bit becomes 0 when OC2FH0 = 1 and writing 1 to the channel 0 overcurrent upper limit exceeded flag 0 clear (DSCOCESCR.CLROC2FH0 / DSCSCRCH0.CLROC2FH). 0: Indicates that the overcurrent data doesn't exceed the upper limit 2 1: Indicates that the overcurrent data has exceeded the upper limit 2	R

Bit	Symbol	Function	R/W
21	OC2FH1	Channel 1 overcurrent upper limit exceeded 2 flag This bit becomes 1 when the overcurrent data is updated and this data is larger than the upper limit 2. This bit becomes 0 when OC2FH1 = 1 and writing 1 to the channel 1 overcurrent upper limit exceeded flag 2 clear (DSCOCESCR.CLROC2FH1 / DSCSCRCH1.CLROC2FH). 0: Indicates that the overcurrent data doesn't exceed the upper limit 2 1: Indicates that the overcurrent data has exceeded the upper limit 2	R
22	OC2FH2	Channel 2 overcurrent upper limit exceeded 2 flag This bit becomes 1 when the overcurrent data is updated and this data is larger than the upper limit 2. This bit becomes 0 when OC2FH2 = 1 and writing 1 to the channel 2 overcurrent upper limit exceeded flag 2 clear (DSCOCESCR.CLROC2FH2 / DSCSCRCH2.CLROC2FH). 0: Indicates that the overcurrent data doesn't exceed the upper limit 2 1: Indicates that the overcurrent data has exceeded the upper limit 2	R
31:23	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

48.2.1.14 DSCOCNSR : Channel Overcurrent Notification Status Register

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x228

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	OWD3 N2	OWD3 N1	OWD3 N0	—	OWD2 N2	OWD2 N1	OWD2 N0	—	OWD1 N2	OWD1 N1	OWD1 N0	—	OWD0 N2	OWD0 N1	OWD0 N0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	OWD0N0	Channel 0 overcurrent detection window notification 0 flag This bit becomes 1 when the condition of overcurrent detection window notification 0 is satisfied. See section 48.3.13.2. Overcurrent Detection Window Notification for detail. This bit becomes 0 when OWD0N0 = 1 and writing 1 to the channel 0 overcurrent detection window notification flag 0 clear (DSCOCNSCR.CLROWD0N0 / DSCSCRCH0.CLROWD0). 0: Indicates that the condition of overcurrent detection window notification 0 is not satisfied 1: Indicates that the condition of overcurrent detection window notification 0 was satisfied	R
1	OWD0N1	Channel 1 overcurrent detection window notification 0 flag This bit becomes 1 when the condition of overcurrent detection window notification 0 is satisfied. See section 48.3.13.2. Overcurrent Detection Window Notification for detail. This bit becomes 0 when OWD0N1 = 1 and writing 1 to the channel 1 overcurrent detection window notification flag 0 clear (DSCOCNSCR.CLROWD0N1 / DSCSCRCH1.CLROWD0). 0: Indicates that the condition of overcurrent detection window notification 0 is not satisfied 1: Indicates that the condition of overcurrent detection window notification 0 was satisfied	R
2	OWD0N2	Channel 2 overcurrent detection window notification 0 flag This bit becomes 1 when the condition of overcurrent detection window notification 0 is satisfied. See section 48.3.13.2. Overcurrent Detection Window Notification for detail. This bit becomes 0 when OWD0N2 = 1 and writing 1 to the channel 2 overcurrent detection window notification flag 0 clear (DSCOCNSCR.CLROWD0N2 / DSCSCRCH2.CLROWD0). 0: Indicates that the condition of overcurrent detection window notification 0 is not satisfied 1: Indicates that the condition of overcurrent detection window notification 0 was satisfied	R

Bit	Symbol	Function	R/W
3	—	This bit is read as 0.	R
4	OWD1N0	Channel 0 overcurrent detection window notification 1 flag This bit becomes 1 when the condition of overcurrent detection window notification 1 is satisfied. See section 48.3.13.2. Overcurrent Detection Window Notification for detail. This bit becomes 0 when OWD1N0 = 1 and writing 1 to the channel 0 overcurrent detection window notification flag 1 clear (DSCOCNSCR.CLROWD1N0 / DSCSCRCH0.CLROWD1). 0: Indicates that the condition of overcurrent detection window notification 1 is not satisfied 1: Indicates that the condition of overcurrent detection window notification 1 was satisfied	R
5	OWD1N1	Channel 1 overcurrent detection window notification 1 flag This bit becomes 1 when the condition of overcurrent detection window notification 1 is satisfied. See section 48.3.13.2. Overcurrent Detection Window Notification for detail. This bit becomes 0 when OWD1N1 = 1 and writing 1 to the channel 1 overcurrent detection window notification flag 1 clear (DSCOCNSCR.CLROWD1N1 / DSCSCRCH1.CLROWD1). 0: Indicates that the condition of overcurrent detection window notification 1 is not satisfied 1: Indicates that the condition of overcurrent detection window notification 1 was satisfied	R
6	OWD1N2	Channel 2 overcurrent detection window notification 1 flag This bit becomes 1 when the condition of overcurrent detection window notification 1 is satisfied. See section 48.3.13.2. Overcurrent Detection Window Notification for detail. This bit becomes 0 when OWD1N2 = 1 and writing 1 to the channel 2 overcurrent detection window notification flag 1 clear (DSCOCNSCR.CLROWD1N2 / DSCSCRCH2.CLROWD1). 0: Indicates that the condition of overcurrent detection window notification 1 is not satisfied 1: Indicates that the condition of overcurrent detection window notification 1 was satisfied	R
7	—	This bit is read as 0.	R
8	OWD2N0	Channel 0 overcurrent detection window notification 2 flag This bit becomes 1 when the condition of overcurrent detection window notification 2 is satisfied. See section 48.3.13.2. Overcurrent Detection Window Notification for detail. This bit becomes 0 when OWD2N0 = 1 and writing 1 to the channel 0 overcurrent detection window notification flag 2 clear (DSCOCNSCR.CLROWD2N0 / DSCSCRCH0.CLROWD2). 0: Indicates that the condition of overcurrent detection window notification 2 is not satisfied 1: Indicates that the condition of overcurrent detection window notification 2 was satisfied	R
9	OWD2N1	Channel 1 overcurrent detection window notification 2 flag This bit becomes 1 when the condition of overcurrent detection window notification 2 is satisfied. See section 48.3.13.2. Overcurrent Detection Window Notification for detail. This bit becomes 0 when OWD2N1 = 1 and writing 1 to the channel 1 overcurrent detection window notification flag 2 clear (DSCOCNSCR.CLROWD2N1 / DSCSCRCH1.CLROWD2). 0: Indicates that the condition of overcurrent detection window notification 2 is not satisfied 1: Indicates that the condition of overcurrent detection window notification 2 was satisfied	R
10	OWD2N2	Channel 2 overcurrent detection window notification 2 flag This bit becomes 1 when the condition of overcurrent detection window notification 2 is satisfied. See section 48.3.13.2. Overcurrent Detection Window Notification for detail. This bit becomes 0 when OWD2N2 = 1 and writing 1 to the channel 2 overcurrent detection window notification flag 2 clear (DSCOCNSCR.CLROWD2N2 / DSCSCRCH2.CLROWD2). 0: Indicates that the condition of overcurrent detection window notification 2 is not satisfied 1: Indicates that the condition of overcurrent detection window notification 2 was satisfied	R
11	—	This bit is read as 0.	R

Bit	Symbol	Function	R/W
12	OWD3N0	Channel 0 overcurrent detection window notification 3 flag This bit becomes 1 when the condition of overcurrent detection window notification 3 is satisfied. See section 48.3.13.2. Overcurrent Detection Window Notification for detail. This bit becomes 0 when OWD3N0 = 1 and writing 1 to the channel 0 overcurrent detection window notification flag 3 clear (DSCOCNSCR.CLROWD3N0 / DSCSCRCH0.CLROWD3). 0: Indicates that the condition of overcurrent detection window notification 3 is not satisfied 1: Indicates that the condition of overcurrent detection window notification 3 was satisfied	R
13	OWD3N1	Channel 1 overcurrent detection window notification 3 flag This bit becomes 1 when the condition of overcurrent detection window notification 3 is satisfied. See section 48.3.13.2. Overcurrent Detection Window Notification for detail. This bit becomes 0 when OWD3N1 = 1 and writing 1 to the channel 1 overcurrent detection window notification flag 3 clear (DSCOCNSCR.CLROWD3N1 / DSCSCRCH1.CLROWD3). 0: Indicates that the condition of overcurrent detection window notification 3 is not satisfied 1: Indicates that the condition of overcurrent detection window notification 3 was satisfied	R
14	OWD3N2	Channel 2 overcurrent detection window notification 3 flag This bit becomes 1 when the condition of overcurrent detection window notification 3 is satisfied. See section 48.3.13.2. Overcurrent Detection Window Notification for detail. This bit becomes 0 when OWD3N2 = 1 and writing 1 to the channel 2 overcurrent detection window notification flag 3 clear (DSCOCNSCR.CLROWD3N2 / DSCSCRCH2.CLROWD3). 0: Indicates that the condition of overcurrent detection window notification 3 is not satisfied 1: Indicates that the condition of overcurrent detection window notification 3 was satisfied	R
31:15	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

48.2.1.15 DSCOCRMR : Channel Overcurrent Comparator Result Monitor Register

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x22C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	OC2C MPH2	OC2C MPH1	OC2C MPH0	—	OC2C MPL2	OC2C MPL1	OC2C MPL0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	OC1C MPH2	OC1C MPH1	OC1C MPH0	—	OC1C MPL2	OC1C MPL1	OC1C MPL0	—	OC0C MPH2	OC0C MPH1	OC0C MPH0	—	OC0C MPL2	OC0C MPL1	OC0C MPL0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	OC0CMPL0	Channel 0 overcurrent detect 0 lower limit compare result 0: Indicates that the overcurrent data is not less than the lower limit 0 1: Indicates that the overcurrent data has been less than the lower limit 0	R
1	OC0CMPL1	Channel 1 overcurrent detect 0 lower limit compare result 0: Indicates that the overcurrent data is not less than the lower limit 0 1: Indicates that the overcurrent data has been less than the lower limit 0	R
2	OC0CMPL2	Channel 2 overcurrent detect 0 lower limit compare result 0: Indicates that the overcurrent data is not less than the lower limit 0 1: Indicates that the overcurrent data has been less than the lower limit 0	R
3	—	This bit is read as 0.	R
4	OC0CMPH0	Channel 0 overcurrent detect 0 upper limit compare result 0: Indicates that the overcurrent data does not exceed the upper limit 0 1: Indicates that the overcurrent data has exceeded the upper limit 0	R

Bit	Symbol	Function	R/W
5	OC0CMPH1	Channel 1 overcurrent detect 0 upper limit compare result 0: Indicates that the overcurrent data does not exceed the upper limit 0 1: Indicates that the overcurrent data has exceeded the upper limit 0	R
6	OC0CMPH2	Channel 2 overcurrent detect 0 upper limit compare result 0: Indicates that the overcurrent data does not exceed the upper limit 0 1: Indicates that the overcurrent data has exceeded the upper limit 0	R
7	—	This bit is read as 0.	R
8	OC1CMPL0	Channel 0 overcurrent detect 1 lower limit compare result 0: Indicates that the overcurrent data is not less than the lower limit 1 1: Indicates that the overcurrent data has been less than the lower limit 1	R
9	OC1CMPL1	Channel 1 overcurrent detect 1 lower limit compare result 0: Indicates that the overcurrent data is not less than the lower limit 1 1: Indicates that the overcurrent data has been less than the lower limit 1	R
10	OC1CMPL2	Channel 2 overcurrent detect 1 lower limit compare result 0: Indicates that the overcurrent data is not less than the lower limit 1 1: Indicates that the overcurrent data has been less than the lower limit 1	R
11	—	This bit is read as 0.	R
12	OC1CMPH0	Channel 0 overcurrent detect 1 upper limit compare result 0: Indicates that the overcurrent data does not exceed the upper limit 1 1: Indicates that the overcurrent data has exceeded the upper limit 1	R
13	OC1CMPH1	Channel 1 overcurrent detect 1 upper limit compare result 0: Indicates that the overcurrent data does not exceed the upper limit 1 1: Indicates that the overcurrent data has exceeded the upper limit 1	R
14	OC1CMPH2	Channel 2 overcurrent detect 1 upper limit compare result 0: Indicates that the overcurrent data does not exceed the upper limit 1 1: Indicates that the overcurrent data has exceeded the upper limit 1	R
15	—	This bit is read as 0.	R
16	OC2CMPL0	Channel 0 overcurrent detect 2 lower limit compare result 0: Indicates that the overcurrent data is not less than the lower limit 2 1: Indicates that the overcurrent data has been less than the lower limit 2	R
17	OC2CMPL1	Channel 1 overcurrent detect 2 lower limit compare result 0: Indicates that the overcurrent data is not less than the lower limit 2 1: Indicates that the overcurrent data has been less than the lower limit 2	R
18	OC2CMPL2	Channel 2 overcurrent detect 2 lower limit compare result 0: Indicates that the overcurrent data is not less than the lower limit 2 1: Indicates that the overcurrent data has been less than the lower limit 2	R
19	—	This bit is read as 0.	R
20	OC2CMPH0	Channel 0 overcurrent detect 2 upper limit compare result 0: Indicates that the overcurrent data does not exceed the upper limit 2 1: Indicates that the overcurrent data has exceeded the upper limit 2	R
21	OC2CMPH1	Channel 1 overcurrent detect 2 upper limit compare result 0: Indicates that the overcurrent data does not exceed the upper limit 2 1: Indicates that the overcurrent data has exceeded the upper limit 2	R
22	OC2CMPH2	Channel 2 overcurrent detect 2 upper limit compare result 0: Indicates that the overcurrent data does not exceed the upper limit 2 1: Indicates that the overcurrent data has exceeded the upper limit 2	R
31:23	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

The DSCOCMR register has monitor bits that return the comparison result with the upper/lower limit table for each overcurrent detection every time the overcurrent data is updated. There is no status clear bit of this register because the result updates when the overcurrent data is updated.

48.2.1.16 DSCSR : Channel Status Register

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x240

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	CBUF 2	CBUF 1	CBUF 0	—	CAUF 2	CAUF 1	CAUF 0	—	DUF2	DUF1	DUF0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DUF0	Channel 0 Data Update flag This bit is the Current Data Register Update Interrupt Status bit. This bit becomes 0 when DUF0 = 1 and 1 to the Channel n Data Update Clear flag (DSCSCR.CLRDUF0/ DSCSCRCH0.CLRDUF) or when the Current Data Register Channel 0 (DSCDRCH0) is read. 0: Indicates that the Current Data register has not been updated 1: Indicates that the Current Data register has been updated	R
1	DUF1	Channel 1 Data Update flag This bit is the Current Data Register Update Interrupt Status bit. This bit becomes 0 when DUF1 = 1 and 1 to the Channel n Data Update Clear flag (DSCSCR.CLRDUF1/ DSCSCRCH1.CLRDUF) or when the Current Data Register Channel 1 (DSCDRCH1) is read. 0: Indicates that the Current Data register has not been updated 1: Indicates that the Current Data register has been updated	R
2	DUF2	Channel 2 Data Update flag This bit is the Current Data Register Update Interrupt Status bit. This bit becomes 0 when DUF2 = 1 and 1 to the Channel n Data Update Clear flag (DSCSCR.CLRDUF2/ DSCSCRCH2.CLRDUF) or when the Current Data Register Channel 2 (DSCDRCH2) is read. 0: Indicates that the Current Data register has not been updated 1: Indicates that the Current Data register has been updated	R
3	—	This bit is read as 0.	R
4	CAUF0	Channel 0 capture data A update flag This bit becomes 0 when CAUF0 = 1 and 1 to the capture data update flag A clear (DSCSCR.CLRCAUF0 / DSCSCRCH0.CLRCAUF) or when the capture current data register A channel 0 (DSCCDRACH0) is read. 0: Indicates that the capture register A has not been updated 1: Indicates that the capture register A has been updated	R
5	CAUF1	Channel 1 capture data A update flag This bit becomes 0 when CAUF1 = 1 and 1 to the capture data update flag A clear (DSCSCR.CLRCAUF1 / DSCSCRCH1.CLRCAUF) or when the capture current data register A channel 1 (DSCCDRACH1) is read. 0: Indicates that the capture register A has not been updated 1: Indicates that the capture register A has been updated	R
6	CAUF2	Channel 2 capture data A update flag This bit becomes 0 when CAUF2 = 1 and 1 to the capture data update flag A clear (DSCSCR.CLRCAUF2 / DSCSCRCH2.CLRCAUF) or when the capture current data register A channel 2 (DSCCDRACH2) is read. 0: Indicates that the capture register A has not been updated 1: Indicates that the capture register A has been updated	R
7	—	This bit is read as 0.	R

Bit	Symbol	Function	R/W
8	CBUF0	Channel 0 capture data B update flag This bit becomes 0 when CBUF0 = 1 and 1 to the capture data update flag B clear (DSCSCR.CLRCBUF0 / DSCESCRCH0.CLRCBUF) or when the capture current data register B channel 0 (DSCCDBCH0) is read. 0: Indicates that the capture register B has not been updated 1: Indicates that the capture register B has been updated	R
9	CBUF1	Channel 1 capture data B update flag This bit becomes 0 when CBUF1 = 1 and 1 to the capture data update flag B clear (DSCSCR.CLRCBUF01 / DSCESCRCH1.CLRCBUF) or when the capture current data register B channel 1 (DSCCDBCH1) is read. 0: Indicates that the capture register B has not been updated 1: Indicates that the capture register B has been updated	R
10	CBUF2	Channel 2 capture data B update flag This bit becomes 0 when CBUF2 = 1 and 1 to the capture data update flag B clear (DSCSCR.CLRCBUF02 / DSCESCRCH2.CLRCBUF) or when the capture current data register B channel 2 (DSCCDBCH2) is read. 0: Indicates that the capture register B has not been updated 1: Indicates that the capture register B has been updated	R
31:11	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

48.2.1.17 DSCSSR : Channel State Status Register

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x244

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	CHST ATE2	—	—	—	CHST ATE1	—	—	—	CHST ATE0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CHSTATE0	Channel 0 state 0: Channel stop 1: Channel in operation	R
3:1	—	These bits are read as 0.	R
4	CHSTATE1	Channel 1 state 0: Channel stop 1: Channel in operation	R
7:5	—	These bits are read as 0.	R
8	CHSTATE2	Channel 2 state 0: Channel stop 1: Channel in operation	R
31:9	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

48.2.1.18 DSCESCR : Channel Error Status Clear Register

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x260

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CLRS UMER RH	CLRS UMER RL
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	CLRS CF2	CLRS CF1	CLRS CF0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CLRSCF0	Channel 0 Overcurrent Lower Limit Detection Flag Clear 0: No effect 1: Clear DSCESR.SCF0 / DSCSRCH0.SCF	W
1	CLRSCF1	Channel 1 Overcurrent Lower Limit Detection Flag Clear 0: No effect 1: Clear DSCESR.SCF1 / DSCSRCH1.SCF	W
2	CLRSCF2	Channel 2 Overcurrent Lower Limit Detection Flag Clear 0: No effect 1: Clear DSCESR.SCF2 / DSCSRCH2.SCF	W
15:3	—	The write value should be 0.	W
16	CLRSUMERRL	Overcurrent Sum Error Lower Limit Detection Flag Clear 0: No effect 1: Clear DSCESR.SUMERRL	W
17	CLRSUMERRH	Overcurrent Sum Error Upper Limit Detection Flag Clear 0: No effect 1: Clear DSCESR.SUMERRH	W
31:18	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

48.2.1.19 DSCOCESCR : Channel Overcurrent Error Status Clear Register

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x264

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	CLRO C2FH2	CLRO C2FH1	CLRO C2FH0	—	CLRO C2FL2	CLRO C2FL1	CLRO C2FL0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	CLRO C1FH2	CLRO C1FH1	CLRO C1FH0	—	CLRO C1FL2	CLRO C1FL1	CLRO C1FL0	—	CLRO C0FH2	CLRO C0FH1	CLRO C0FH0	—	CLRO C0FL2	CLRO C0FL1	CLRO C0FL0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CLROC0FL0	Channel 0 overcurrent lower limit detection 0 flag clear 0: No effect 1: Clear DSCOCESR.OC0FL0 / DSCSRCH0.OC0FL	W

Bit	Symbol	Function	R/W
1	CLROC0FL1	Channel 1 overcurrent lower limit detection 0 flag clear 0: No effect 1: Clear DSCOCESR.OC0FL1 / DSCSRCH1.OC0FL	W
2	CLROC0FL2	Channel 2 overcurrent lower limit detection 0 flag clear 0: No effect 1: Clear DSCOCESR.OC0FL2 / DSCSRCH2.OC0FL	W
3	—	The write value should be 0.	W
4	CLROC0FH0	Channel 0 overcurrent upper limit exceeded 0 flag clear 0: No effect 1: Clear DSCOCESR.OC0FH0 / DSCSRCH0.OC0FH	W
5	CLROC0FH1	Channel 1 overcurrent upper limit exceeded 0 flag clear 0: No effect 1: Clear DSCOCESR.OC0FH1 / DSCSRCH1.OC0FH	W
6	CLROC0FH2	Channel 2 overcurrent upper limit exceeded 0 flag clear 0: No effect 1: Clear DSCOCESR.OC0FH2 / DSCSRCH2.OC0FH	W
7	—	The write value should be 0.	W
8	CLROC1FL0	Channel 0 overcurrent lower limit detection 1 flag clear 0: No effect 1: Clear DSCOCESR.OC1FL0 / DSCSRCH0.OC1FL	W
9	CLROC1FL1	Channel 1 overcurrent lower limit detection 1 flag clear 0: No effect 1: Clear DSCOCESR.OC1FL1 / DSCSRCH1.OC1FL	W
10	CLROC1FL2	Channel 2 overcurrent lower limit detection 1 flag clear 0: No effect 1: Clear DSCOCESR.OC1FL2 / DSCSRCH2.OC1FL	W
11	—	The write value should be 0.	W
12	CLROC1FH0	Channel 0 overcurrent upper limit exceeded 1 flag clear 0: No effect 1: Clear DSCOCESR.OC1FH0 / DSCSRCH0.OC1FH	W
13	CLROC1FH1	Channel 1 overcurrent upper limit exceeded 1 flag clear 0: No effect 1: Clear DSCOCESR.OC1FH1 / DSCSRCH1.OC1FH	W
14	CLROC1FH2	Channel 2 overcurrent upper limit exceeded 1 flag clear 0: No effect 1: Clear DSCOCESR.OC1FH2 / DSCSRCH2.OC1FH	W
15	—	The write value should be 0.	W
16	CLROC2FL0	Channel 0 overcurrent lower limit detection 2 flag clear 0: No effect 1: Clear DSCOCESR.OC2FL0 / DSCSRCH0.OC2FL	W
17	CLROC2FL1	Channel 1 overcurrent lower limit detection 2 flag clear 0: No effect 1: Clear DSCOCESR.OC2FL1 / DSCSRCH1.OC2FL	W
18	CLROC2FL2	Channel 2 overcurrent lower limit detection 2 flag clear 0: No effect 1: Clear DSCOCESR.OC2FL2 / DSCSRCH2.OC2FL	W
19	—	The write value should be 0.	W
20	CLROC2FH0	Channel 0 overcurrent upper limit exceeded 2 flag clear 0: No effect 1: Clear DSCOCESR.OC2FH0 / DSCSRCH0.OC2FH	W
21	CLROC2FH1	Channel 1 overcurrent upper limit exceeded 2 flag 0: No effect 1: Clear DSCOCESR.OC2FH1 / DSCSRCH1.OC2FH	W

Bit	Symbol	Function	R/W
22	CLROC2FH2	Channel 2 overcurrent upper limit exceeded 2 flag clear 0: No effect 1: Clear DSCOCESR.OC2FH2 / DSCSRCH2.OC2FH	W
31:23	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

48.2.1.20 DSCOCNSCR : Channel Overcurrent Notification Status Clear Register

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x268

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	CLRO WD3N 2	CLRO WD3N 1	CLRO WD3N 0	—	CLRO WD2N 2	CLRO WD2N 1	CLRO WD2N 0	—	CLRO WD1N 2	CLRO WD1N 1	CLRO WD1N 0	—	CLRO WD0N 2	CLRO WD0N 1	CLRO WD0N 0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CLROWD0N0	Channel 0 overcurrent detection window notification 0 flag clear 0: No effect 1: Clear DSCOCNSR.OWD0N0 / DSCSRCH0.OWD0N	W
1	CLROWD0N1	Channel 1 overcurrent detection window notification 0 flag clear 0: No effect 1: Clear DSCOCNSR.OWD0N1 / DSCSRCH1.OWD0N	W
2	CLROWD0N2	Channel 2 overcurrent detection window notification 0 flag clear 0: No effect 1: Clear DSCOCNSR.OWD0N2 / DSCSRCH2.OWD0N	W
3	—	The write value should be 0.	W
4	CLROWD1N0	Channel 0 overcurrent detection window notification 1 flag clear 0: No effect 1: Clear DSCOCNSR.OWD1N0 / DSCSRCH0.OWD1N	W
5	CLROWD1N1	Channel 1 overcurrent detection window notification 1 flag clear 0: No effect 1: Clear DSCOCNSR.OWD1N1 / DSCSRCH1.OWD1N	W
6	CLROWD1N2	Channel 2 overcurrent detection window notification 1 flag clear 0: No effect 1: Clear DSCOCNSR.OWD1N2 / DSCSRCH2.OWD1N	W
7	—	The write value should be 0.	W
8	CLROWD2N0	Channel 0 overcurrent detection window notification 2 flag clear 0: No effect 1: Clear DSCOCNSR.OWD2N0 / DSCSRCH0.OWD2N	W
9	CLROWD2N1	Channel 1 overcurrent detection window notification 2 flag clear 0: No effect 1: Clear DSCOCNSR.OWD2N1 / DSCSRCH1.OWD2N	W
10	CLROWD2N2	Channel 2 overcurrent detection window notification 2 flag clear 0: No effect 1: Clear DSCOCNSR.OWD2N2 / DSCSRCH2.OWD2N	W
11	—	The write value should be 0.	W

Bit	Symbol	Function	R/W
12	CLROWD3N0	Channel 0 overcurrent detection window notification 3 flag clear 0: No effect 1: Clear DSCOCNSR.OWD3N0 / DSCSRCH0.OWD3N	W
13	CLROWD3N1	Channel 1 overcurrent detection window notification 3 flag clear 0: No effect 1: Clear DSCOCNSR.OWD3N1 / DSCSRCH1.OWD3N	W
14	CLROWD3N2	Channel 2 overcurrent detection window notification 3 flag clear 0: No effect 1: Clear DSCOCNSR.OWD3N2 / DSCSRCH2.OWD3N	W
31:15	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

48.2.1.21 DSCSCR : Channel Status Clear Register

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x280

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	CLRC BUF2	CLRC BUF1	CLRC BUF0	—	CLRC AUF2	CLRC AUF1	CLRC AUF0	—	CLRD UF2	CLRD UF1	CLRD UF0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CLRDUF0	Channel 0 Data Update Flag Clear 0: No effect 1: Clear DSCSR.DUF0 / DSCSRCH0.DUF	W
1	CLRDUF1	Channel 1 Data Update Flag Clear 0: No effect 1: Clear DSCSR.DUF1 / DSCSRCH1.DUF	W
2	CLRDUF2	Channel 2 Data Update Flag Clear 0: No effect 1: Clear DSCSR.DUF2 / DSCSRCH2.DUF	W
3	—	The write value should be 0.	W
4	CLRCAUF0	Channel 0 Capture Data A Update Flag Clear 0: No effect 1: Clear DSCSR.CAUF0 / DSCSRCH0.CAUF	W
5	CLRCAUF1	Channel 1 Capture Data A Update Flag Clear 0: No effect 1: Clear DSCSR.CAUF1 / DSCSRCH1.CAUF	W
6	CLRCAUF2	Channel 2 Capture Data A Update Flag Clear 0: No effect 1: Clear DSCSR.CAUF2 / DSCSRCH2.CAUF	W
7	—	The write value should be 0.	W
8	CLRCBUF0	Channel 0 Capture Data B Update Flag Clear 0: No effect 1: Clear DSCSR.CBUF0 / DSCSRCH0.CBUF	W
9	CLRCBUF1	Channel 1 Capture Data B Update Flag Clear 0: No effect 1: Clear DSCSR.CBUF1 / DSCSRCH1.CBUF	W

Bit	Symbol	Function	R/W
10	CLRCBUF2	Channel 2 Capture Data B Update Flag Clear 0: No effect 1: Clear DSCSR.CBUF2 / DSCSRCH2.CBUF	W
31:11	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

48.2.2 DSMIF Channel Control Registers

48.2.2.1 DSICRCHn : Interrupt Control Register Channel n (n = 0 to 2)

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x300 + 0x140 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	OWNE 3	OWNE 2	OWNE 1	OWNE 0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	IOEH2	IOEL2	IOEH1	IOEL1	IOEH0	IOEL0	—	—	—	—	ISE	IBUE	IAUE	IUE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	IUE	Current data register update interrupt enable 0: Do not output current data register update interrupt 1: Current data register update interrupt can be output	R/W
1	IAUE	Capture current data register A update interrupt enable 0: Do not output capture current data register A update interrupt 1: Capture current data register A update interrupt can be output	R/W
2	IBUE	Capture current data register B update interrupt enable 0: Do not output capture current data register B update interrupt 1: Capture current data register B update interrupt can be output	R/W
3	ISE	Short circuit detection error interrupt enable bit 0: Do not output short circuit detection interrupt 1: Short circuit detection interrupt can be output	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W
8	IOEL0	Overcurrent lower limit detection interrupt 0 0: Do not output the overcurrent lower limit detection interrupt 0 1: Overcurrent lower limit detection interrupt 0 can be output	R/W
9	IOEH0	Overcurrent upper limit exceeded detection interrupt 0 0: Do not output the overcurrent upper limit detection interrupt 0 1: Overcurrent upper limit exceeded detection interrupt 0 can be output	R/W
10	IOEL1	Overcurrent lower limit detection interrupt 1 0: Do not output the overcurrent lower limit detection interrupt 1 1: Overcurrent lower limit detection interrupt 1 can be output	R/W
11	IOEH1	Overcurrent upper limit exceeded detection interrupt 1 0: Do not output the overcurrent upper limit detection interrupt 1 1: Overcurrent upper limit exceeded detection interrupt 1 can be output	R/W
12	IOEL2	Overcurrent lower limit detection interrupt 2 0: Do not output the overcurrent lower limit detection interrupt 2 1: Overcurrent lower limit detection interrupt 2 can be output	R/W
13	IOEH2	Overcurrent upper limit exceeded detection interrupt 2 0: Do not output the overcurrent upper limit detection interrupt 2 1: Overcurrent upper limit exceeded detection interrupt 2 can be output	R/W

Bit	Symbol	Function	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W
16	OWNE0	Overcurrent detection window notification 0 output enable 0: Do not output overcurrent detection window notification 0 1: Overcurrent detection window notification 0 can be output	R/W
17	OWNE1	Overcurrent detection window notification 1 output enable 0: Do not output overcurrent detection window notification 1 1: Overcurrent detection window notification 1 can be output	R/W
18	OWNE2	Overcurrent detection window notification 2 output enable 0: Do not output overcurrent detection window notification 2 1: Overcurrent detection window notification 2 can be output	R/W
19	OWNE3	Overcurrent detection window notification 3 output enable 0: Do not output overcurrent detection window notification 3 1: Overcurrent detection window notification 3 can be output	R/W
31:20	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.2.2 DSCMCCRCHn : Current Measurement Clock Control Register Channel n (n = 0 to 2)

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x304 + 0x140 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	CKDIV[5:0]						SEDE	—	—	—	—	—	—	CKDIR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CKDIR	A/D conversion clock master/slave switching 0: DSMCLKn pin is input (slave operation) 1: DSMCLKn pin is output (master operation)	R/W
6:1	—	These bits are read as 0. The write value should be 0.	R/W
7	SEDE	Sampling edge selection 0: Capture DSMDATAN at the negative edge of DSMCLKn 1: Capture DSMDATAN at the positive edge of DSMCLKn	R/W
13:8	CKDIV[5:0]	A/D conversion clock division ratio Set the division ratio in clock master mode. Set to a division ratio of $2 \times (\text{CKDIV}[5:0] + 1)$ with the core clock	R/W
31:14	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.2.3 DSCMFCRCHn : Current Measurement Filter Control Register Channel n (n = 0 to 2)

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x308 + 0x140 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	CMSH[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	CMDEC[7:0]								—	—	—	—	—	—	CMSINC[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	CMSINC[1:0]	Current measurement filter order setting 0 0: Sinc3 (3rd order) 0 1: Sinc1 (1st order) 1 0: Sinc2 (2nd order) 1 1: Setting prohibited	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
15:8	CMDEC[7:0]	Decimation ratio selection for current measurement M indicates the decimation ratio. Decimation ratio M is set as follows: M = CMDEC + 1 The decimation ratio M can be set within the following range. 4 ≤ M ≤ 256 (0x3 ≤ CMDEC ≤ 0xFF) Setting less than M = 4 is prohibited. When these value is set, Sinc filter operation is not guaranteed.	R/W
20:16	CMSH[4:0]	Data shift setting for current measurement*1 0x00: [23:8] 0x01: [22:7] 0x02: [21:6] 0x03: [20:5] 0x04: [19:4] 0x05: [18:3] 0x06: [17:2] 0x07: [16:1] 0x08: [15:0] 0x09: [14:0]<<1 0x0A: [13:0]<<2 0x0B: [12:0]<<3 0x0C: [11:0]<<4 0x0D: [10:0]<<5 0x0E: [9:0]<<6 0x0F: [8:0]<<7 0x10: [7:0]<<8 0x11: [6:0]<<9 0x12: [5:0]<<10 0x13: [4:0]<<11 Others: Setting prohibited	R/W
31:21	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. In CMSH[4:0] = 0x09 to 0x13, the lower bits are filled with 0s.

48.2.2.4 DSCMCTCRCHn : Current Measurement Capture Trigger Control Register Channel n (n = 0 to 2)

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x30C + 0x140 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	DEDE E	—	—	—	—	—	DITSEL[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	CTSELB[2:0]			—	—	—	—	—	CTSELA[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	CTSELA[2:0]	Current capture trigger A selection bit 0x0: Do not capture 0x1: Select current data capture trigger 0 0x2: Select current data capture trigger 1 0x3: Select current data capture trigger 2 0x4: Select current data capture trigger 3 0x5: Select current data capture trigger 4 0x6: Select current data capture trigger 5 0x7: Setting prohibited (do not capture)	R/W
7:3	—	These bits are read as 0. The write value should be 0.	R/W
10:8	CTSELB[2:0]	Current capture trigger B selection bit 0x0: Do not capture 0x1: Select current data capture trigger 0 0x2: Select current data capture trigger 1 0x3: Select current data capture trigger 2 0x4: Select current data capture trigger 3 0x5: Select current data capture trigger 4 0x6: Select current data capture trigger 5 0x7: Setting prohibited (do not capture)	R/W
15:11	—	These bits are read as 0. The write value should be 0.	R/W
17:16	DITSEL[1:0]	Current measurement filter initialization trigger selection bit for frequency division counter for decimation. 0 0: Do not initialize 0 1: Select decimation dividing counter initialization trigger 0 1 0: Select decimation dividing counter initialization trigger 1 1 1: Select decimation dividing counter initialization trigger 2	R/W
22:18	—	These bits are read as 0. The write value should be 0.	R/W
23	DEDEGE	Current measurement filter initialization trigger for division counter for decimation edge selection bit. The trigger from ELC is usually used positive edge. Change from the initial value if necessary. 0: The frequency division counter for decimation is initialized at the negative edge of initialization trigger 1: The frequency division counter for decimation is initialized at the positive edge of initialization trigger	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Base address: $\text{DSMIFm} = 0x4032_A000 + 0x0800 \times m$ ($m = 0, 1$)

Offset address: $0x320 + 0x140 \times n$

Bit position: 31

Bit field:

0

Value after reset: 0

Note: S-TYPE-3, P-TYPE-3

Base address: $\text{DSMIFm} = 0x4032_A000 + 0x0800 \times m$ ($m = 0, 1$)
Offset address: $0x330 + 0x140 \times n$

Bit position: 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Bit field:

31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

OCSH[4:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
---------------	----	----	----	----	----	----	---	---	---	---	---	---	---	---	---	---

Bit field:

OCDEC[7:0]

—	OCSINC[1:0]
---	-------------

[illegible]

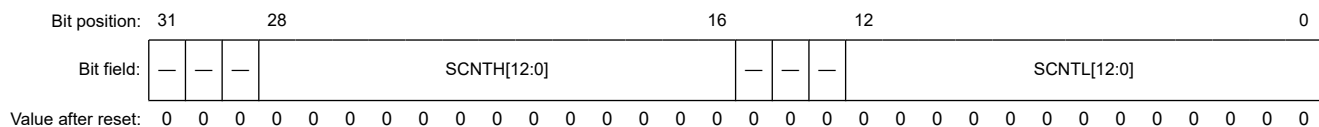
Bit	Symbol	Function	R/W
1:0	OCSINC[1:0]	Overcurrent detection filter order setting 0 0: Sinc3 (3rd order) 0 1: Sinc1 (1st order) 1 0: Sinc2 (2nd order) 1 1: Setting prohibited	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
15:8	OCDEC[7:0]	Decimation ratio selection for overcurrent detection M indicates the decimation ratio. Decimation ratio M is set as follows: $M = OCDEC + 1$ The decimation ratio M can be set within the following range. $4 \leq M \leq 256$ ($0x3 \leq OCDEC \leq 0xFF$) Setting less than $M = 4$ is prohibited. When these value is set, Sinc filter operation is not guaranteed.	R/W

Bit	Symbol	Function	R/W
20:16	OCSH[4:0]	Data shift setting for overcurrent detection*1 0x00: [23:8] 0x01: [22:7] 0x02: [21:6] 0x03: [20:5] 0x04: [19:4] 0x05: [18:3] 0x06: [17:2] 0x07: [16:1] 0x08: [15:0] 0x09: [14:0]<<1 0x0A: [13:0]<<2 0x0B: [12:0]<<3 0x0C: [11:0]<<4 0x0D: [10:0]<<5 0x0E: [9:0]<<6 0x0F: [8:0]<<7 0x10: [7:0]<<8 0x11: [6:0]<<9 0x12: [5:0]<<10 0x13: [4:0]<<11 Others: Setting prohibited	R/W
31:21	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. In OCSH[4:0] = 0x09 to 0x13, the lower bits are filled with 0s.

48.2.2.7 DSSCTSRCHn : Short Circuit Threshold Setting Register Channel n (n = 0 to 2)

Offset address: $0x324 + 0x140 \times n$



Bit	Symbol	Function	R/W
12:0	SCNTL[12:0]	Short circuit detection low continuous detection count Set the upper limit for the number of times 0 is input continuously to the DSMDATAN pin.	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
28:16	SCNTH[12:0]	Short circuit detection high continuous detection count Set the upper limit for the number of times 1 is input continuously to the DSMDATAN pin.	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.2.8 DSODCRCHn : Overcurrent Detect Control Register Channel n (n = 0 to 2)

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x334 + 0x140 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	OWFE 3	OWFE 2	OWFE 1	OWFE 0	—	—	ODEH 2	ODEL 2	ODEH 1	ODEL 1	ODEH 0	ODEL 0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ODEL0	Overcurrent lower limit detection 0 enable 0: Overcurrent lower limit detection 0 is invalid 1: Overcurrent lower limit detection 0 is valid	R/W
1	ODEH0	Overcurrent upper limit exceeded detection 0 enable 0: Overcurrent upper limit exceeded detection 0 is invalid 1: Overcurrent upper limit exceeded detection 0 is valid	R/W
2	ODEL1	Overcurrent lower limit detection 1 enable 0: Overcurrent lower limit detection 1 is invalid 1: Overcurrent lower limit detection 1 is valid	R/W
3	ODEH1	Overcurrent upper limit exceeded detection 1 enable 0: Overcurrent upper limit exceeded detection 1 is invalid 1: Overcurrent upper limit exceeded detection 1 is valid	R/W
4	ODEL2	Overcurrent lower limit detection 2 enable 0: Overcurrent lower limit detection 2 is invalid 1: Overcurrent lower limit detection 2 is valid	R/W
5	ODEH2	Overcurrent upper limit exceeded detection 2 enable 0: Overcurrent upper limit exceeded detection 2 is invalid 1: Overcurrent upper limit exceeded detection 2 is valid	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W
8	OWFE0	Overcurrent lower limit detection interrupt 0 0: Do not output the overcurrent lower limit detection interrupt 0 1: Overcurrent lower limit detection interrupt 0 can be output	R/W
9	OWFE1	Overcurrent detection window function 0 enable 0: Overcurrent detection window function 0 is invalid 1: Overcurrent detection window function 0 is valid	R/W
10	OWFE2	Overcurrent detection window function 1 enable 0: Overcurrent detection window function 1 is invalid 1: Overcurrent detection window function 1 is valid	R/W
11	OWFE3	Overcurrent detection window function 2 enable 0: Overcurrent detection window function 2 is invalid 1: Overcurrent detection window function 2 is valid	R/W
31:12	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.2.9 DSODWCRCHn : Overcurrent Detect Window Control Register Channel n (n = 0 to 2)

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x338 + 0x140 × n

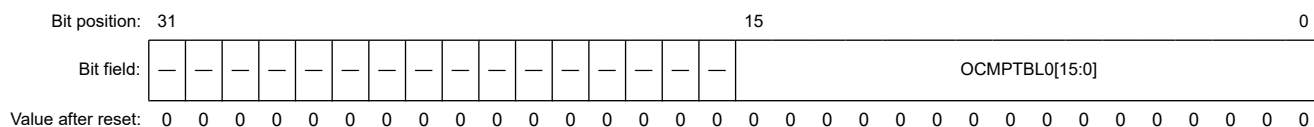
Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	OWNM3[3:0]				OWN M2	OWN M1	OWN M0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	OWNM0	Channel n overcurrent detection window notification 0 mode select 0: (overcurrent data channel n < lower limit 0) or (overcurrent data channel n > upper limit 0) 1: lower limit 0 channel n ≤ overcurrent data channel n ≤ upper limit 0 channel n	R/W
1	OWNM1	Channel n overcurrent detection window notification 1 mode select 0: (overcurrent data channel n < lower limit 1) or (overcurrent data channel n > upper limit 1) 1: lower limit 1 channel n ≤ overcurrent data channel n ≤ upper limit 1 channel n	R/W
2	OWNM2	Channel n overcurrent detection window notification 2 mode select 0: (overcurrent data channel n < lower limit 2) or (overcurrent data channel n > upper limit 2) 1: lower limit 2 channel n ≤ overcurrent data channel n ≤ upper limit 2 channel n	R/W
6:3	OWNM3[3:0]	Channel n overcurrent detection window notification 3 mode select 0x0: Overcurrent detection window notification 0 0x1: Overcurrent detection window notification 1 0x2: Overcurrent detection window notification 2 0x3: Overcurrent detection window notification 0 OR overcurrent detection window notification 1 0x4: Overcurrent detection window notification 0 AND overcurrent detection window notification 1 0x5: Overcurrent detection window notification 0 OR overcurrent detection window notification 2 0x6: Overcurrent detection window notification 0 AND overcurrent detection window notification 2 0x7: Overcurrent detection window notification 1 OR overcurrent detection window notification 2 0x8: Overcurrent detection window notification 1 AND overcurrent detection window notification 2 0x9: Overcurrent detection window notification 0 OR overcurrent detection window notification 1 OR overcurrent detection window notification 2 0xA: Overcurrent detection window notification 0 AND overcurrent detection window notification 1 AND overcurrent detection window notification 2 Others: Setting prohibited	R/W
31:7	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Base address: $\text{DSMIFm} = 0x4032_A000 + 0x0800 \times m$ ($m = 0, 1$)

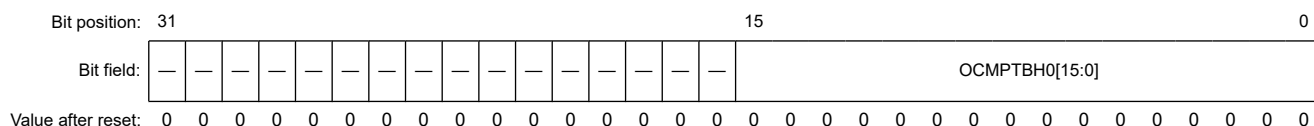
Offset address: $0x3A0 + 0x140 \times n$



Bit	Symbol	Function	R/W
15:0	OCMPTBL0[15:0]	Overcurrent detection lower limit 0 This bit sets the lower limit of the active current value for overcurrent detection 0.	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

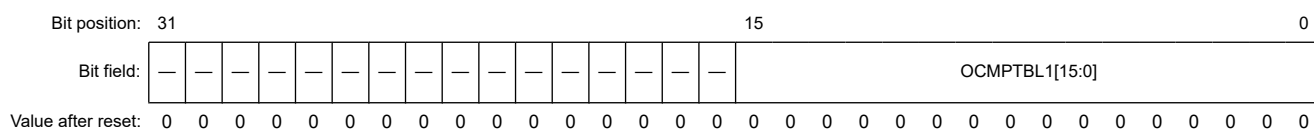
Base address: $\text{DSMIFm} = 0x4032_A000 + 0x0800 \times m$ ($m = 0, 1$)
Offset address: $0x3A4 + 0x140 \times n$



Bit	Symbol	Function	R/W
15:0	OCMPTBH0[15:0]	Overcurrent detection upper limit 0 This bit sets the upper limit of the active current value for overcurrent detection 0.	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Base address: $\text{DSMIFm} = 0x4032_A000 + 0x0800 \times m$ ($m = 0, 1$)
Offset address: $0x3A8 + 0x140 \times n$



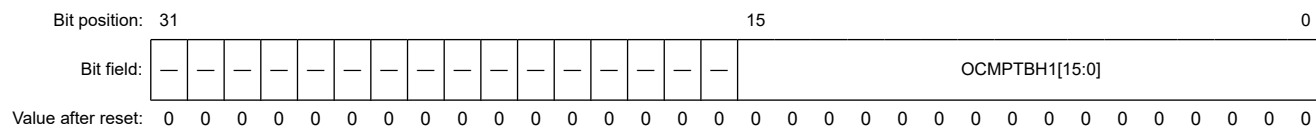
Bit	Symbol	Function	R/W
15:0	OCMPTBL1[15:0]	Overcurrent detection lower limit 1 This bit sets the lower limit of the active current value for overcurrent detection 1.	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.2.13 DSOCHTR1CHn : Overcurrent High Threshold Register 1 Channel n (n = 0 to 2)

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x3AC + 0x140 × n



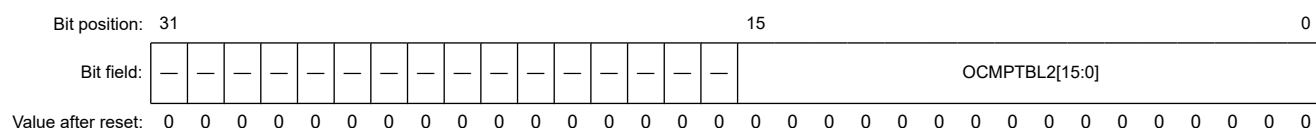
Bit	Symbol	Function	R/W
15:0	OCMPTBH1[15:0]	Overcurrent detection upper limit 1 This bit sets the upper limit of the active current value for overcurrent detection 1.	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.2.14 DSOCLTR2CHn : Overcurrent Low Threshold Register 2 Channel n (n = 0 to 2)

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x3B0 + 0x140 × n



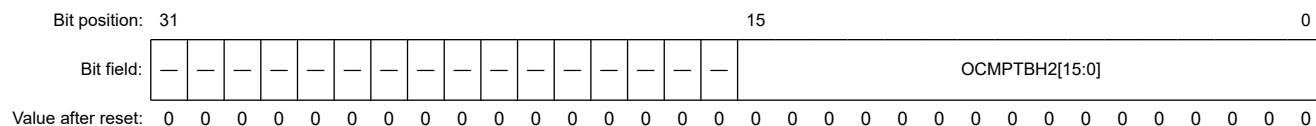
Bit	Symbol	Function	R/W
15:0	OCMPTBL2[15:0]	Overcurrent detection lower limit 2 This bit sets the lower limit of the active current value for overcurrent detection 2.	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.2.15 DSOCHTR2CHn : Overcurrent High Threshold Register 2 Channel n (n = 0 to 2)

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x3B4 + 0x140 × n



Bit	Symbol	Function	R/W
15:0	OCMPTBH2[15:0]	Overcurrent detection upper limit 2 This bit sets the upper limit of the active current value for overcurrent detection 2.	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.2.16 DSCSTRTRCHn : Software Start Trigger Register Channel n (n = 0 to 2)

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x3E0 + 0x140 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	STRT RG
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	STRTRG	Channel start trigger Mirror register of DSCSTRTR.STRTRGn 0: Do nothing 1: Start channel n	W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.2.17 DSCSTPTRCHn : Software Stop Trigger Register Channel n (n = 0 to 2)

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x3E4 + 0x140 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	STPT RG
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	STPTRG	Channel stop trigger Mirror register of DSCSTRTR.STPTRGn 0: Do nothing 1: Stop channel n	W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

48.2.2.18 DSCDRCHn : Current Data Register Channel n (n = 0 to 2)

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x3F0 + 0x140 × n

Bit position:	31															15																	0
Bit field:																ADDR[15:0]																	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		

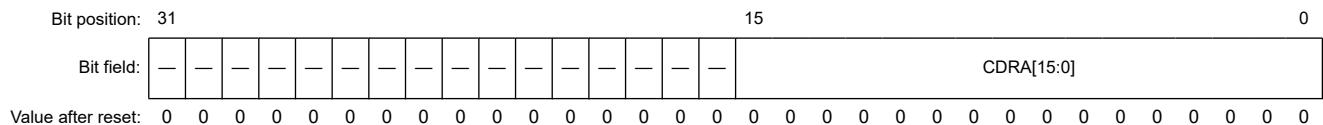
Bit	Symbol	Function	R/W
15:0	ADDR[15:0]	Current data Stores the current value generated by the current measurement data block.	R
31:16	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

48.2.2.19 DSCCDRACHn : Capture Current Data Register A Channel n (n = 0 to 2)

Base address: $DSMIF_m = 0x4032_A000 + 0x0800 \times m$ ($m = 0, 1$)

Offset address: $0x3F4 + 0x140 \times n$



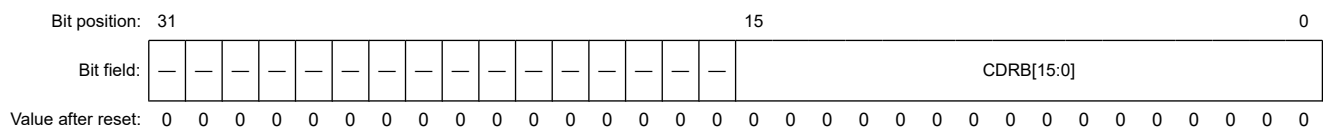
Bit	Symbol	Function	R/W
15:0	CDRA[15:0]	Capture current data A The current data value when the capture trigger A is input is stored.	R
31:16	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

48.2.2.20 DSCDRBCHn : Capture Current Data Register B Channel n (n = 0 to 2)

Base address: $DSMIF_m = 0x4032_A000 + 0x0800 \times m$ ($m = 0, 1$)

Offset address: $0x3F8 + 0x140 \times n$



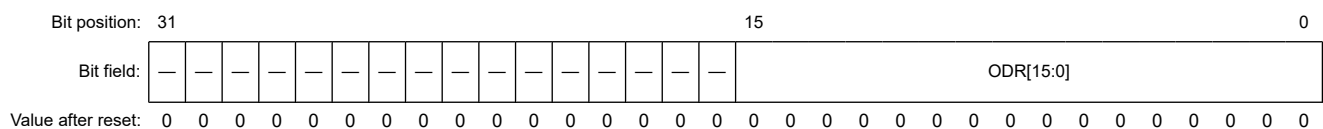
Bit	Symbol	Function	R/W
15:0	CDRB[15:0]	Capture current data B The current data value when the capture trigger B is input is stored.	R
31:16	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

48.2.2.21 DSOCDRCHn : Overcurrent Data Register Channel n (n = 0 to 2)

Base address: $\text{DSMIFm} = 0x4032_A000 + 0x0800 \times m$ ($m = 0, 1$)

Offset address: $0x3FC + 0x140 \times n$



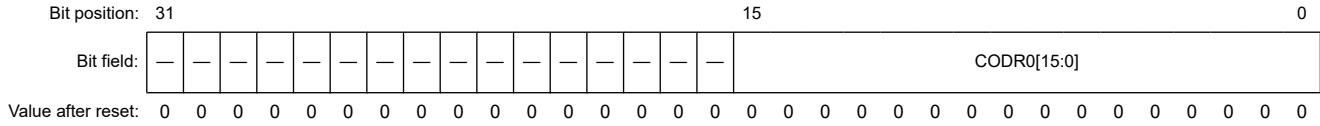
Bit	Symbol	Function	R/W
15:0	ODR[15:0]	Overcurrent data Stores the current value generated by the overcurrent data block.	R
31:16	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

48.2.2.22 DSCOCDR0CHn : Capture Overcurrent Data Register 0 Channel n (n = 0 to 2)

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x400 + 0x140 × n



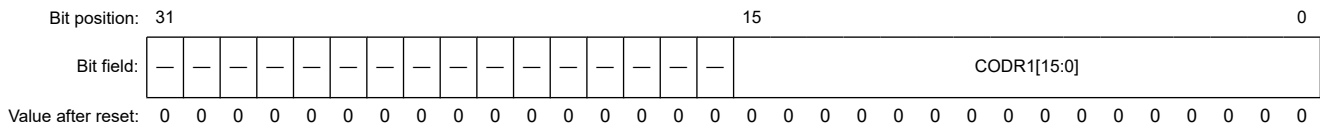
Bit	Symbol	Function	R/W
15:0	CODR0[15:0]	Capture overcurrent data 0 When either the overcurrent upper limit or the overcurrent lower limit is detected by the overcurrent data block, the overcurrent data 0 used for comparison is stored.	R
31:16	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

48.2.2.23 DSCOCDR1CHn : Capture Overcurrent Data Register 1 Channel n (n = 0 to 2)

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x404 + 0x140 × n



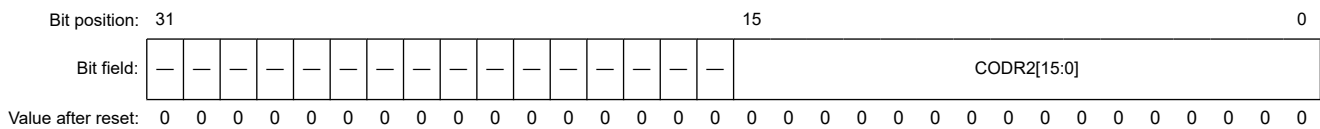
Bit	Symbol	Function	R/W
15:0	CODR1[15:0]	Capture overcurrent data 1 When either the overcurrent upper limit or the overcurrent lower limit is detected by the overcurrent data block, the overcurrent data 1 used for comparison is stored.	R
31:16	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

48.2.2.24 DSCOCDR2CHn : Capture Overcurrent Data Register 2 Channel n (n = 0 to 2)

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x408 + 0x140 × n



Bit	Symbol	Function	R/W
15:0	CODR2[15:0]	Capture overcurrent data 2 When either the overcurrent upper limit or the overcurrent lower limit is detected by the overcurrent data block, the overcurrent data 2 used for comparison is stored.	R
31:16	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

48.2.2.25 DSCSRCHn : Status Register Channel n (n = 0 to 2)

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x420 + 0x140 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	OC2C MPH	OC2C MPL	OC1C MPH	OC1C MPL	OC0C MPH	OC0C MPL	—	—	—	—	OWD3 N	OWD2 N	OWD1 N	OWD0 N
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	OC2F H	OC2F L	OC1F H	OC1F L	OC0F H	OC0F L	CHSTATE	—	—	—	SCF	CBUF	CAUF	DUF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DUF	Channel n data update flag Mirror register of DSCSR.DUFLn 0: Indicates that the current data register has not been updated 1: Indicates that the current data register has been updated	R
1	CAUF	Channel n capture data A update flag Mirror register of DSCSR.CAUFn 0: Indicates that the capture data register A has not been updated 1: Indicates that the capture data register A has been updated	R
2	CBUF	Channel n capture data B update flag Mirror register of DSCSR.CBUFn 0: Indicates that the capture data register B has not been updated 1: Indicates that the capture data register B has been updated	R
3	SCF	Channel n short circuit detection flag Mirror register of DSCSR.SCFn 0: Indicates that the DSMDATAN pin short circuit is not detected 1: Indicates that a short circuit on the DSMDATAN pin has been detected	R
6:4	—	These bits are read as 0.	R
7	CHSTATE	Channel n state Mirror register of DSCSR.CHSTATEn 0: Channel stop 1: Channel in operation	R
8	OC0FL	Channel n overcurrent lower limit detection 0 flag Mirror register of DSCSR.OC0FLn 0: Indicates that the overcurrent data is not less than the lower limit 0 1: Indicates that the overcurrent data has been less than the lower limit 0	R
9	OC0FH	Channel n overcurrent upper limit exceeded 0 flag Mirror register of DSCSR.OC0Fhn 0: Indicates that the overcurrent data doesn't exceed the upper limit 0 1: Indicates that the overcurrent data has exceeded the upper limit 0	R
10	OC1FL	Channel n overcurrent lower limit detection 1 flag Mirror register of DSCSR.OC1FLn 0: Indicates that the overcurrent data is not less than the lower limit 1 1: Indicates that the overcurrent data has been less than the lower limit 1	R
11	OC1FH	Channel n overcurrent upper limit exceeded 1 flag Mirror register of DSCSR.OC1Fhn 0: Indicates that the overcurrent data doesn't exceed the upper limit 1 1: Indicates that the overcurrent data has exceeded the upper limit 1	R
12	OC2FL	Channel n overcurrent lower limit detection 2 flag Mirror register of DSCSR.OC2FLn 0: Indicates that the overcurrent data is not less than the lower limit 2 1: Indicates that the overcurrent data has been less than the lower limit 2	R

Bit	Symbol	Function	R/W
13	OC2FH	Channel n overcurrent upper limit exceeded 2 flag Mirror register of DSCOCESR.OC2FHz 0: Indicates that the overcurrent data doesn't exceed the upper limit 2 1: Indicates that the overcurrent data has exceeded the upper limit 2	R
15:14	—	These bits are read as 0.	R
16	OWD0N	Channel n overcurrent detection window notification 0 Mirror register of DSCOCNSR.OWD0Nz 0: Indicates that the condition of overcurrent detection window notification 0 is not satisfied 1: Indicates that the condition of overcurrent detection window notification 0 was satisfied	R
17	OWD1N	Channel n overcurrent detection window notification 1 Mirror register of DSCOCNSR.OWD1Nz 0: Indicates that the condition of overcurrent detection window notification 1 is not satisfied 1: Indicates that the condition of overcurrent detection window notification 1 was satisfied	R
18	OWD2N	Channel n overcurrent detection window notification 2 Mirror register of DSCOCNSR.OWD2Nz 0: Indicates that the condition of overcurrent detection window notification 2 is not satisfied 1: Indicates that the condition of overcurrent detection window notification 2 was satisfied	R
19	OWD3N	Channel n overcurrent detection window notification 3 Mirror register of DSCOCNSR.OWD3Nz 0: Indicates that the condition of overcurrent detection window notification 3 is not satisfied 1: Indicates that the condition of overcurrent detection window notification 3 was satisfied	R
23:20	—	These bits are read as 0.	R
24	OC0CMPL	Channel n overcurrent detect 0 lower limit compare result Mirror register of DSCOCMR.OC0CMPLz 0: Indicates that the overcurrent data is not less than the lower limit 0 1: Indicates that the overcurrent data has fallen the lower limit 0	R
25	OC0CMPH	Channel n overcurrent detect 0 upper limit compare result Mirror register of DSCOCMR.OC0CMPHz 0: Indicates that the overcurrent data doesn't exceed the upper limit 0 1: Indicates that the overcurrent data has exceeded the upper limit 0	R
26	OC1CMPL	Channel n overcurrent detect 1 lower limit compare result Mirror register of DSCOCMR.OC1CMPLz 0: Indicates that the overcurrent data is not less than the lower limit 1 1: Indicates that the overcurrent data has fallen the lower limit 1	R
27	OC1CMPH	Channel n overcurrent detect 1 upper limit compare result Mirror register of DSCOCMR.OC1CMPHz 0: Indicates that the overcurrent data doesn't exceed the upper limit 1 1: Indicates that the overcurrent data has exceeded the upper limit 1	R
28	OC2CMPL	Channel n overcurrent detect 2 lower limit compare result Mirror register of DSCOCMR.OC2CMPLz 0: Indicates that the overcurrent data is not less than the lower limit 2 1: Indicates that the overcurrent data has fallen the lower limit 2	R
29	OC2CMPH	Channel n overcurrent detect 2 upper limit compare result Mirror register of DSCOCMR.OC2CMPHz 0: Indicates that the overcurrent data doesn't exceed the upper limit 2 1: Indicates that the overcurrent data has exceeded the upper limit 2	R
31:30	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

The **DSCSRCHn** register is a mirror register of various registers in [section 48.2.1. DSMIF Common Control Registers](#).

48.2.2.26 DSCSCRCHn : Status Clear Register Channel n (n = 0 to 2)

Base address: DSMIFm = 0x4032_A000 + 0x0800 × m (m = 0, 1)

Offset address: 0x430 + 0x140 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	CLRO WD3N	CLRO WD2N	CLRO WD1N	CLRO WD0N
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	CLRO C2FH	CLRO C2FL	CLRO C1FH	CLRO C1FL	CLRO C0FH	CLRO C0FL	—	—	—	—	CLRS CF	CLRC BUF	CLRC AUF	CLRD UF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CLRDUF	Channel n data update flag clear Mirror register of DSCSCR.CLRDUFn 0: Do nothing 1: Clear DSCSCR.DUFn / DSCSRCHn.DUF	W
1	CLRCAUF	Channel n capture data A update flag clear Mirror register of DSCSCR.CLRCAUFn 0: Do nothing 1: Clear DSCSCR.CAUFn / DSCSRCHn.CAUF	W
2	CLRCBUF	Channel n capture data B update flag clear Mirror register of DSCSCR.CLRCBUFn 0: Do nothing 1: Clear DSCSCR.CBUFn / DSCSRCHn.CBUF	W
3	CLRSCF	Channel n short circuit detection flag clear Mirror register of DSCSCR.CLRSCFn 0: Do nothing 1: Clear DSCSCR.SCFn / DSCSRCHn.SCF	W
7:4	—	The write value should be 0.	W
8	CLROC0FL	Channel n overcurrent lower limit detection flag 0 clear Mirror register of DSCOCESR.CLROC0FLn 0: Do nothing 1: Clear DSCOCESR.OSC0FLn / DSCSRCHn.OC0FL	W
9	CLROC0FH	Channel n overcurrent upper limit exceeded flag 0 clear Mirror register of DSCOCESR.CLROC0FHn 0: Do nothing 1: Clear DSCOCESR.OSC0FHn / DSCSRCHn.OC0FH	W
10	CLROC1FL	Channel n overcurrent lower limit detection flag 1 clear Mirror register of DSCOCESR.CLROC1FLn 0: Do nothing 1: Clear DSCOCESR.OSC1FLn / DSCSRCHn.OC1FL	W
11	CLROC1FH	Channel n overcurrent upper limit exceeded flag 1 clear Mirror register of DSCOCESR.CLROC1FHn 0: Do nothing 1: Clear DSCOCESR.OSC1FHn / DSCSRCHn.OC1FH	W
12	CLROC2FL	Channel n overcurrent lower limit detection flag 2 clear Mirror register of DSCOCESR.CLROC2FLn 0: Do nothing 1: Clear DSCOCESR.OSC2FLn / DSCSRCHn.OC2FL	W
13	CLROC2FH	Channel n overcurrent upper limit exceeded flag 2 clear Mirror register of DSCOCESR.CLROC2FHn 0: Do nothing 1: Clear DSCOCESR.OSC2FHn / DSCSRCHn.OC2FH	W
15:14	—	The write value should be 0.	W

Bit	Symbol	Function	R/W
16	CLROWD0N	Channel n overcurrent detection window notification 0 flag clear Mirror register of DSCOCNSCR.CLROWD0n 0: Do nothing 1: Clear DSCOCNSR.OWD0Nn / DSCSRCHn.OWD0N	W
17	CLROWD1N	Channel n overcurrent detection window notification 1 flag clear Mirror register of DSCOCNSCR.CLROWD1n 0: Do nothing 1: Clear DSCOCNSR.OWD1Nn / DSCSRCHn.OWD1N	W
18	CLROWD2N	Channel n overcurrent detection window notification 2 flag clear Mirror register of DSCOCNSCR.CLROWD2n 0: Do nothing 1: Clear DSCOCNSR.OWD2Nn / DSCSRCHn.OWD2N	W
19	CLROWD3N	Channel n overcurrent detection window notification 3 flag clear Mirror register of DSCOCNSCR.CLROWD3n 0: Do nothing 1: Clear DSCOCNSR.OWD3Nn / DSCSRCHn.OWD3N	W
31:20	—	The write value should be 0.	W

Note: S-TYPE-3, P-TYPE-3

The **DSCSRCHn** register is a mirror register of various registers in [section 48.2.1. DSMIF Common Control Registers](#).

48.3 Operation

48.3.1 Start Flow

[Figure 48.2](#) shows an example of a start procedure for DSMIF.

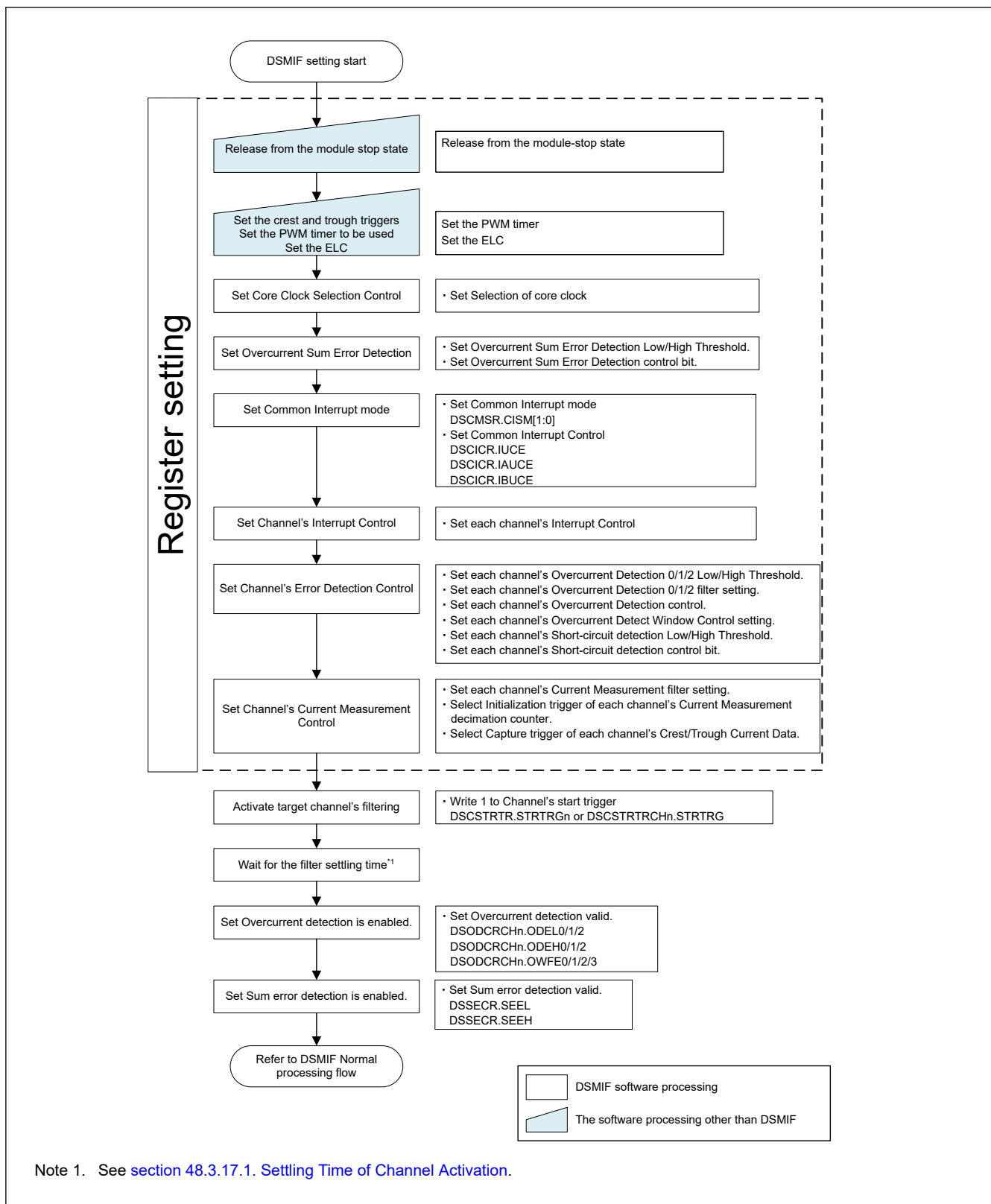


Figure 48.2 Start flow of DSMIF

48.3.2 Normal Processing Flow

Figure 48.3 shows an example of a normal processing procedure for DSMIF.

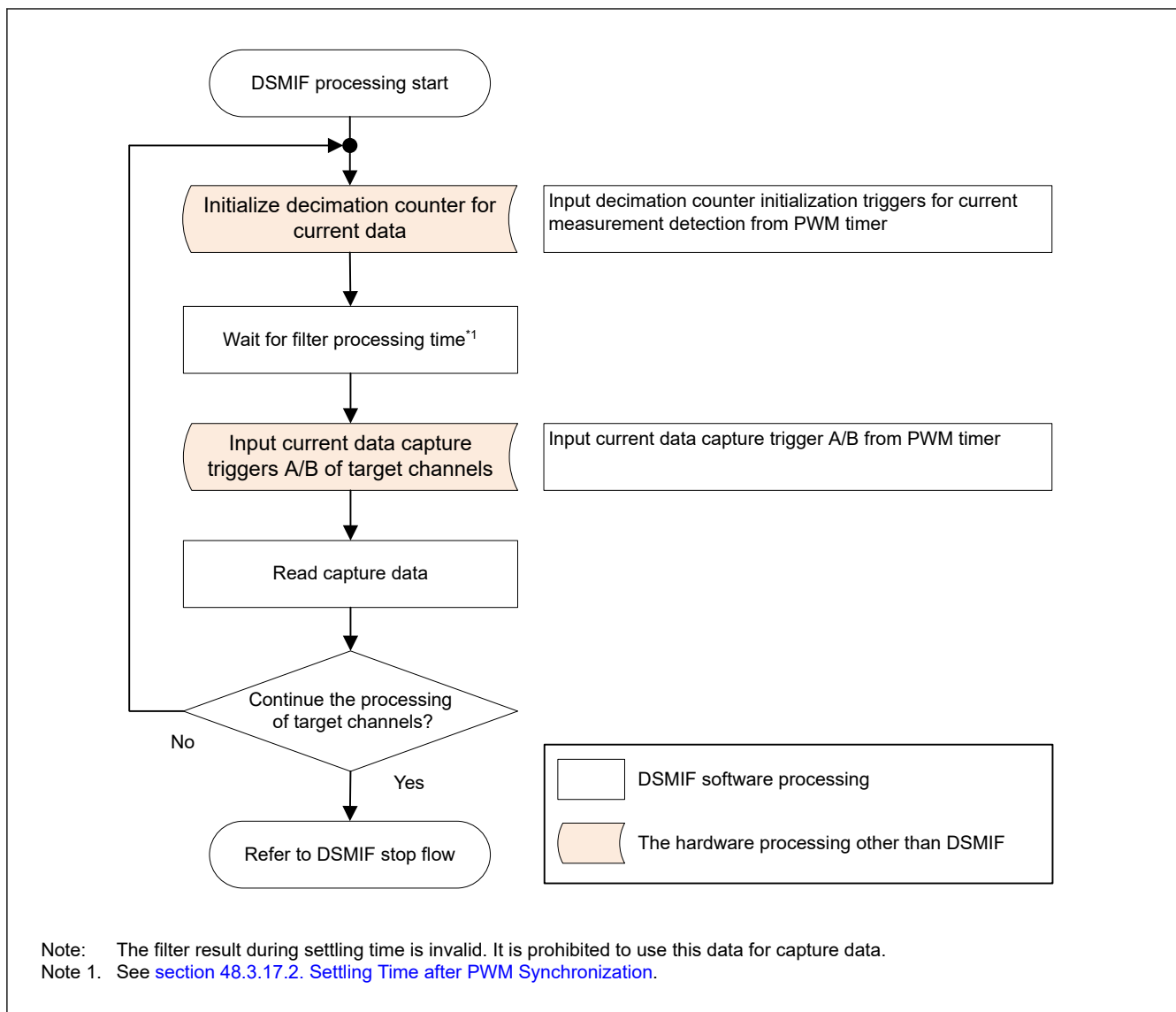


Figure 48.3 Normal processing flow for DSMIF

48.3.3 Stop Flow

[Figure 48.4](#) shows an example of a stop procedure in channel synchronization control mode for DSMIF. [Figure 48.5](#) shows the example of the stop procedure in channel individual control mode of DSMIF.

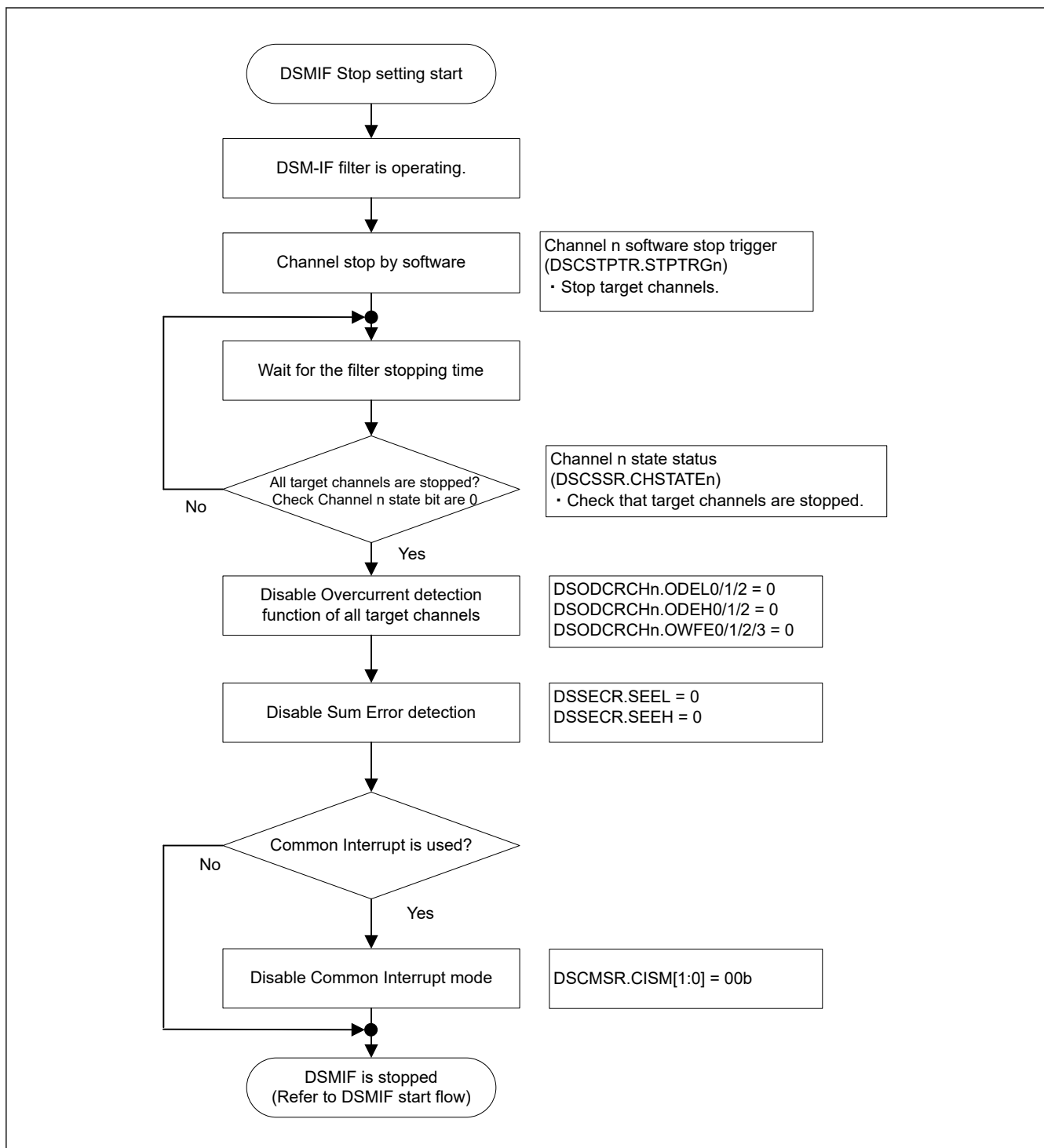


Figure 48.4 Stop flow for DSMIF (Channel synchronization control mode)

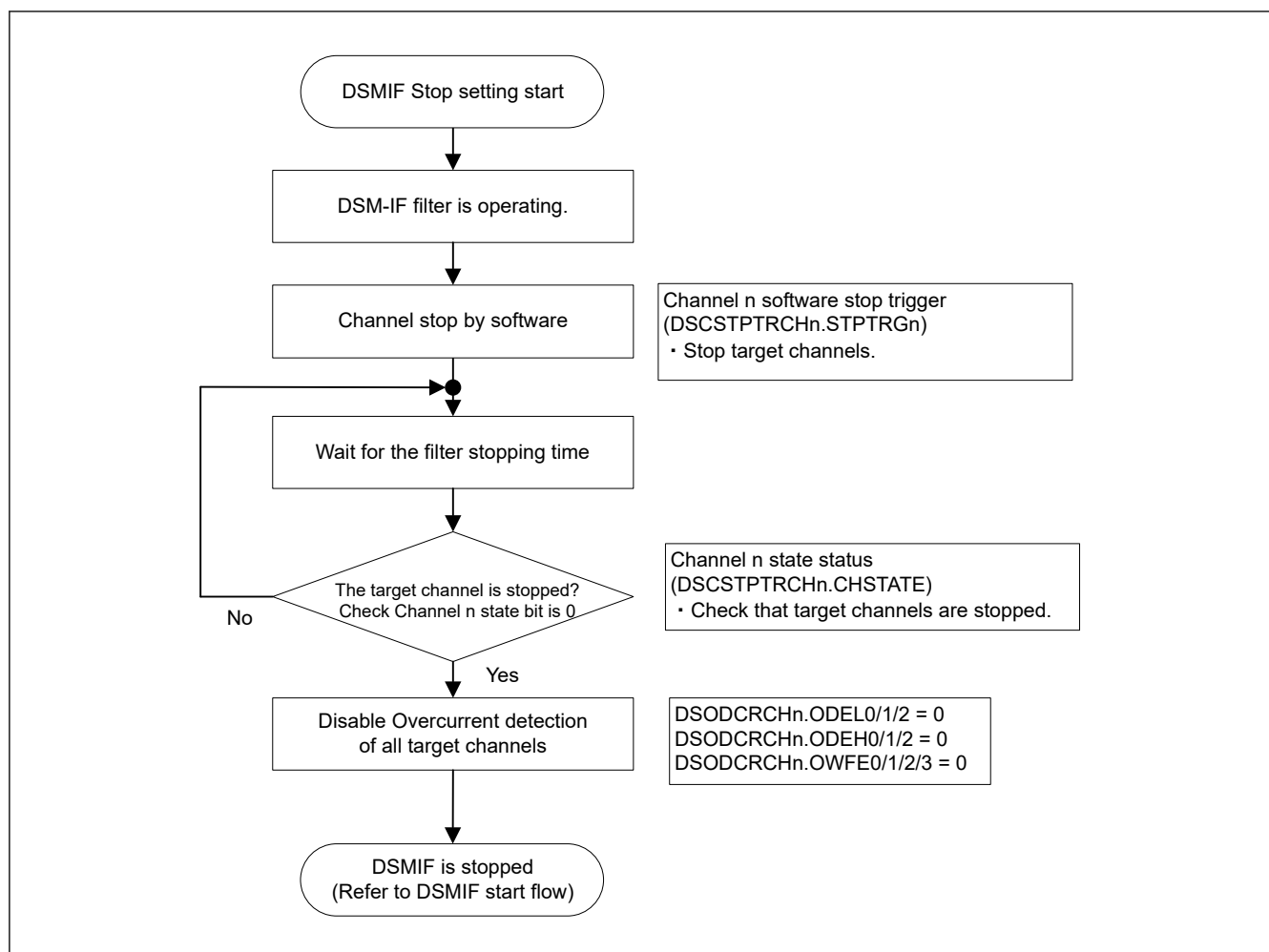


Figure 48.5 Stop flow for DSMIF (Channel individual control mode)

48.3.4 Error Flow

Figure 48.6 shows an example of an error procedure for DSMIF.

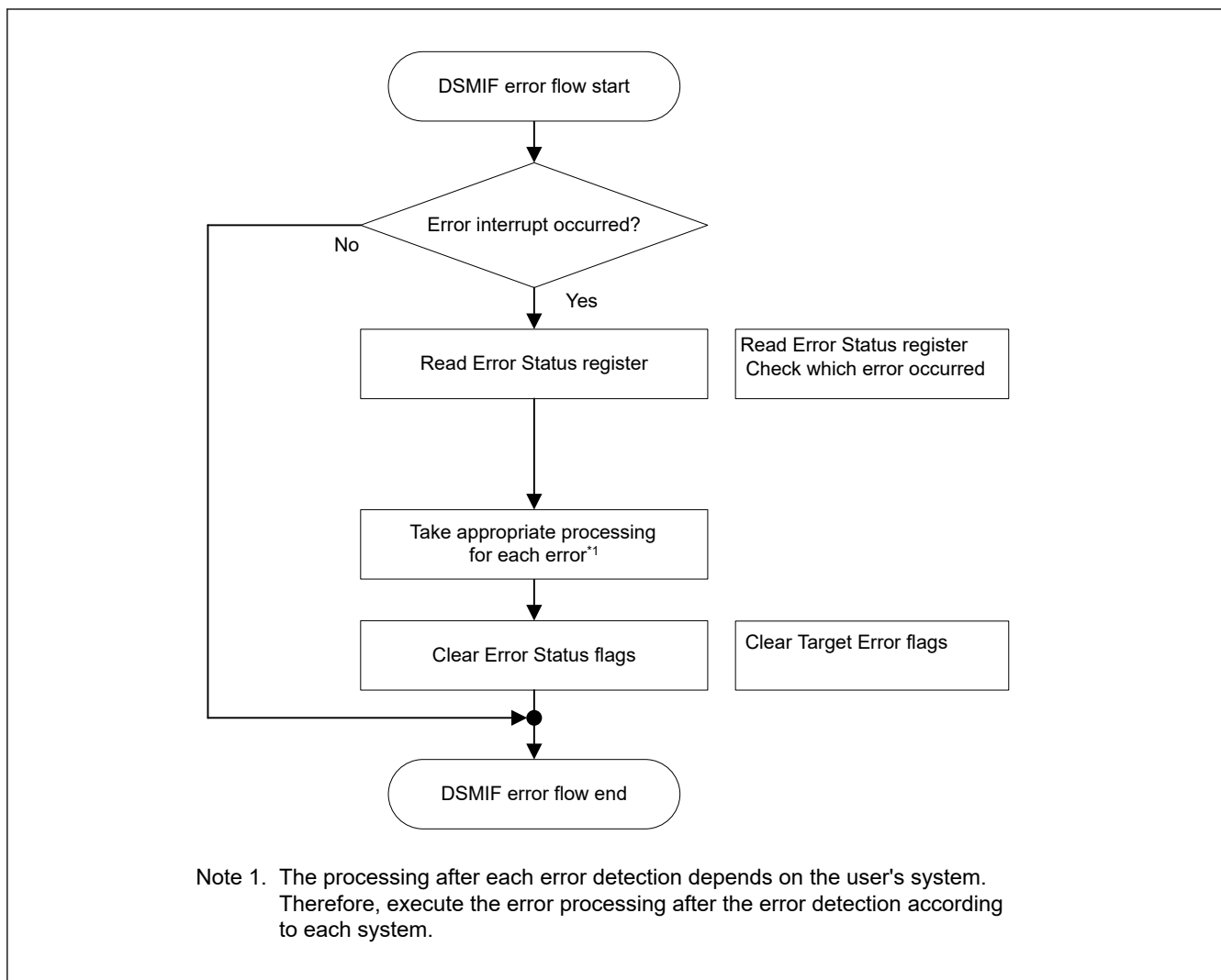


Figure 48.6 Error flow for DSMIF

48.3.5 Clock Specification

When DSCMCCRCHn.CKDIR is set to 0 (clock slave mode), an external delta-sigma modulator clock is input to DSMCLKn. When DSCMCCRCHn.CKDIR is set to 1 (clock master mode), the frequency-divided core clock is output to DSMCLKn.

The frequency of the DSMCLKn can be selected by the DSCMCCRCHn.CKDIV[5:0] bits.

Table 48.3 shows the clock division ratio of master mode in DSMIF.

Table 48.3 Clock division ratio example of master mode in DSMIF (1 of 2)

No.	Core clock (MHz)	DSMCLKn (MHz)	Supported core clock division ratio	Setting value (DSCMCCRCHn.CKDIV[5:0])
6	300	25	12	0x05
7		16.67	18	0x08
8		15	20	0x09
9		10	30	0x0E
10		5	60	0x1D

Table 48.3 Clock division ratio example of master mode in DSMIF (2 of 2)

No.	Core clock (MHz)	DSMCLKn (MHz)	Supported core clock division ratio	Setting value (DSCMCCRCHn.CKDIV[5:0])
11	200	25	8	0x03
12		20	10	0x04
13		12.5	16	0x07
14		10	20	0x09
15		6.25	32	0x0F
16		5	40	0x13

48.3.6 Channel Start and Channel Stop

The channel start and channel stop are controlled by the register in [Table 48.4](#). Set the channels to be synchronized to 1 simultaneously. Set the channel start according to [section 48.3.1. Start Flow](#). Set the channel stop according to [section 48.3.3. Stop Flow](#).

Table 48.4 Channel start trigger and channel stop trigger

Channel mode	Category	Register bit name
Channel synchronization control mode	Channel start	DSCSTRTR.STRTRG0
		DSCSTRTR.STRTRG1
		DSCSTRTR.STRTRG2
	Channel stop	DSCSTPTR.STPTRG0
		DSCSTPTR.STPTRG1
		DSCSTPTR.STPTRG2
Channel individual control mode	Channel start	DSCSTRTRCHn.STRTRG
	Channel stop	DSCSTPTRCHn.STPTRG

48.3.7 Channel Synchronization

To synchronize the Sinc filters of multiple channels, use the following procedure. To allow for multiple channels to be synchronized, follow the procedure in [section 48.3.1. Start Flow](#).

1. Set the same clock settings.
2. Select the same trigger for division counter initialization trigger.
Set the same value to Division counter initialization trigger select bits (DSCMCTCRCHn.DITSEL[1:0]).
3. Select the same trigger for capture trigger A and B.
 - (a) Set the same value to the Current Capture Trigger A Select bits (DSCMCTCRCHn.CTSELA[2:0]).
 - (b) Set the same value to the Current Capture Trigger B Select bits (DSCMCTCRCHn.CTSELB[2:0]).
4. Activate the channels simultaneously
Write 1 to Channel n start trigger (DSCSTRTR.STRTRGn) at the same time.

48.3.8 Data Input Control

By using the DSCMCCRCHn.SEDGE bit, data receive can be selected to receive on the falling edge of DSMCLKn or the rising edge of DSMCLKn. Data input control receives clock and data from DSM. Synchronized data and clock are output to the sinc filter and short-circuit detection.

48.3.9 Sinc Filter

The differentiation stage is operated by decimation clock (the frequency is 1/M of DSMCLKn). M is the decimation ratio and is set in the DSCMFCRCHn.CMDEC[7:0] bits for current measurement or in the DSOCFCRCHn.OCDEC[7:0] bits for overcurrent detection. The filter result is stored in register every decimation clock. In the case of current data, an interrupt is output at the same time when the filter result is stored.

Figure 48.7 shows a block diagram for the Sinc Filter.

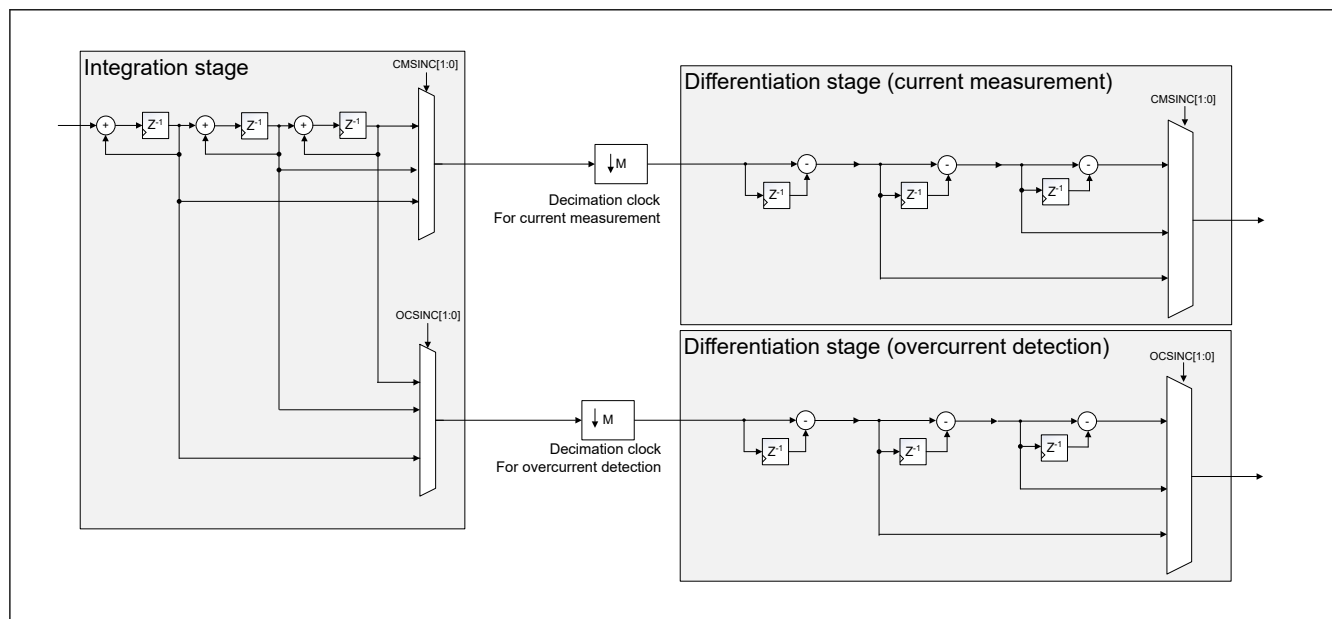


Figure 48.7 Block diagram of Sinc filter

48.3.10 Available Settings of Sinc Filter

Table 48.5 lists the available settings of the sinc filter and data shifting.

Table 48.5 Available settings of Sinc filter (1 of 2)

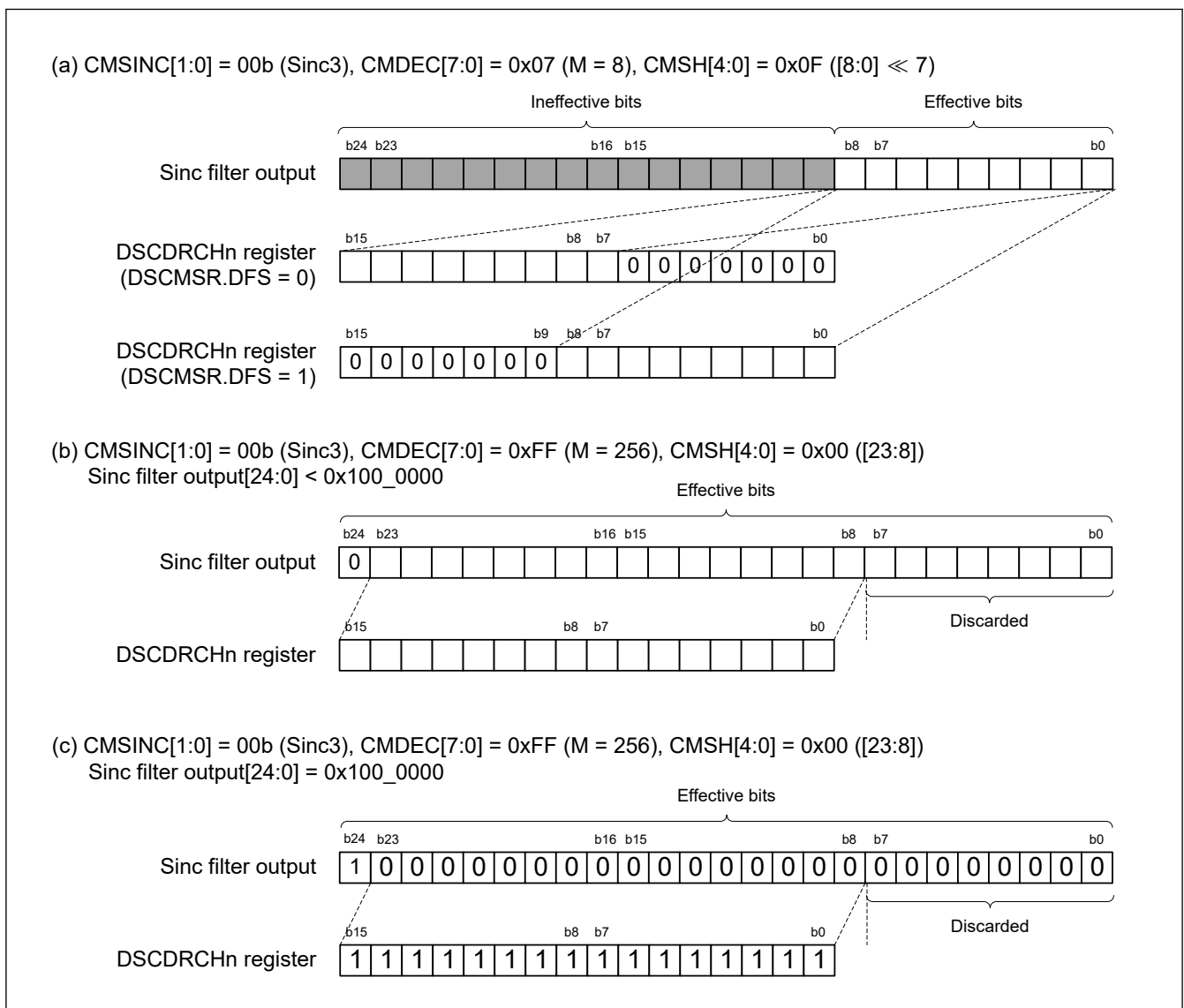
Sinc filter order	Decimation rate	Effective bits	Peak value (Decimal)	Peak value (Hexadecimal)	Stored data		CMSH[4:0] OCSH[4:0]	Resolution
					DFS = 0	DFS = 1		
3 (Sinc3)	4	[6:0]	63 ^{*1}	000003F	[5:0], 00_0000_0000	00_0000_0000, [5:0]	0x12	6 bits
	5	[6:0]	125	000007D	[6:0], 0_0000_0000	0_0000_0000, [6:0]	0x11	7 bits
	6	[7:0]	216	00000D8	[7:0], 0000_0000	0000_0000, [7:0]	0x10	8 bits
	7	[8:0]	343	0000157	[8:0], 000_0000	000_0000, [8:0]	0x0F	9 bits
	8	[9:0]	511 ^{*1}	00001FF	[8:0], 000_0000	000_0000, [8:0]	0x0F	9 bits
	9-10	[9:0]	1000	00003E8	[9:0], 00_0000	00_0000, [9:0]	0x0E	10 bits
	11-12	[10:0]	1728	00006C0	[10:0], 0_0000	0_0000, [10:0]	0x0D	11 bits
	13-15	[11:0]	3375	0000D2F	[11:0], 0000	0000, [11:0]	0x0C	12 bits
	16	[12:0]	4095 ^{*1}	0000FFF	[11:0], 0000	0000, [11:0]	0x0C	12 bits
	17-20	[12:0]	8000	0001F40	[12:0], 000	000, [12:0]	0x0B	13 bits
	21-25	[13:0]	15625	0003D09	[13:0], 00	00, [13:0]	0x0A	14 bits
	26-31	[14:0]	29791	000745F	[14:0], 0	0, [14:0]	0x09	15 bits
	32	[15:0]	32767 ^{*1}	0007FFF	[14:0], 0	0, [14:0]	0x09	15 bits
	33-40	[15:0]	64000	000FA00	[15:0]		0x08	16 bits
	41-50	[16:0]	125000	001E848	[16:1]		0x07	16 bits
	51-63	[17:0]	250047	003D0BF	[17:2]		0x06	16 bits
	64	[18:0]	262143 ^{*1}	003FFFF	[17:2]		0x06	16 bits
	65-80	[18:0]	512000	007D000	[18:3]		0x05	16 bits
	81-101	[19:0]	1030301	00FB89D	[19:4]		0x04	16 bits
	102-127	[20:0]	2048383	01F417F	[20:5]		0x03	16 bits
	128	[21:0]	2097151 ^{*1}	01FFFFFF	[20:5]		0x03	16 bits
	129-161	[21:0]	4173281	03FADE1	[21:6]		0x02	16 bits
	162-203	[22:0]	8365427	07FA573	[22:7]		0x01	16 bits
	204-255	[23:0]	16581375	0FD02FF	[23:8]		0x00	16 bits
	256	[24:0]	16777215 ^{*1}	0FFFFFFF	[23:8]		0x00	16 bits
2 (Sinc2)	16	[7:0]	255 ^{*1}	00000FF	[7:0], 0000_0000	0000_0000, [7:0]	0x10	8 bits
	17-20	[8:0]	400	0000190	[8:0], 000_0000	000_0000, [8:0]	0x0F	9 bits
	21-31	[9:0]	961	00003C1	[9:0], 00_0000	00_0000, [9:0]	0x0E	10 bits
	32	[10:0]	1023 ^{*1}	00003FF	[9:0], 00_0000	00_0000, [9:0]	0x0E	10 bits
	33-40	[10:0]	1600	0000640	[10:0], 0_0000	0_0000, [10:0]	0x0D	11 bits
	41-63	[11:0]	3969	0000F81	[11:0], 0000	0000, [11:0]	0x0C	12 bits
	64	[12:0]	4095 ^{*1}	0000FFF	[11:0], 0000	0000, [11:0]	0x0C	12 bits
	65-90	[12:0]	8100	0001FA4	[12:0], 000	000, [12:0]	0x0B	13 bits
	91-127	[13:0]	16129	0003F01	[13:0], 00	00, [13:0]	0x0A	14 bits
	128	[14:0]	16383 ^{*1}	0003FFF	[13:0], 00	00, [13:0]	0x0A	14 bits
	129-181	[14:0]	32761	0007FF9	[14:0], 0	0, [14:0]	0x09	15 bits
	182-255	[15:0]	65025	000FE01	[15:0]		0x08	16 bits
	256	[16:0]	65535 ^{*1}	000FFFF	[15:0]		0x08	16 bits

Table 48.5 Available settings of Sinc filter (2 of 2)

Sinc filter order	Decimation rate	Effective bits	Peak value (Decimal)	Peak value (Hexadecimal)	Stored data		CM SH[4:0] OCSH[4:0]	Resolution
					DFS = 0	DFS = 1		
1 (Sinc1)	32	[5:0]	31 ^{*1}	000001F	[4:0], 000_0000_0000	000_0000_0000, [4:0]	0x13	5 bits
	33-63	[5:0]	63	000003F	[5:0], 00_0000_0000	00_0000_0000, [5:0]	0x12	6 bits
	64	[6:0]	63 ^{*1}	000003F	[5:0], 00_0000_0000	00_0000_0000, [5:0]	0x12	6 bits
	65-127	[6:0]	127	000007F	[6:0], 0_0000_0000	0_0000_0000, [6:0]	0x11	7 bits
	128	[7:0]	127 ^{*1}	000007F	[6:0], 0_0000_0000	0_0000_0000, [6:0]	0x11	7 bits
	129-255	[7:0]	255	00000FF	[7:0], 0000_0000	0000_0000, [7:0]	0x10	8 bits
	256	[8:0]	255 ^{*1}	00000FF	[7:0], 0000_0000	0000_0000, [7:0]	0x10	8 bits

Note 1. When the filter result is the maximum value, it is rounded to the maximum value of -1.

Figure 48.8 shows the relationship between the filter settings and the stored values.

**Figure 48.8 Relationship between the filter settings and the stored values**

48.3.11 Current Measurement Data Processing

Each channel of DSMIF has the data registers specified in [Table 48.6](#). Capture trigger A is selected by DSCMCTCRCHn.CTSELA[2:0] and capture trigger B is selected by DSCMCTCRCHn.CTSELB[2:0].

The trigger capturing function allows the capture of the values for current at the time of capture trigger A and capture trigger B using the event link controller (ELC) to set the capture trigger A or capture trigger B.

Table 48.6 Current measurement data list

No.	Register name	Symbol	Stored data
1	Current Data register	DSCDRCHn	The measured values for current
2	Capture Current Data Register A	DSCCDRACHn	The captured value of Current Data register by capture trigger A from the internal timer module
3	Capture Current Data Register B	DSCCDRBCHn	The captured value of Current Data register by capture trigger B from the internal timer module

48.3.12 Short-circuit Detection

When the short-circuit detection enable (DSEDCRCHn.SDE) is valid, the short-circuit detection can operate. A dedicated 13-bit counter counts consecutive 0's or 1's input from dsmdata n (n = 0 to 2). When the number of consecutive 0's exceeds the value set in the DSSCTSRCHn.SCNTL[12:0] register or the number of consecutive 1's exceeds the value set in the DSSCTSRCHn.SCNTL[12:0] register, a short-circuit detection interrupt request is generated.

48.3.13 Overcurrent Detection

The overcurrent detection function is used for abnormally detection. Overcurrent data is generated by the sinc filter result that is setting in [section 48.3.9. Sinc Filter](#). The overcurrent data is compared with the value of the upper limit value or the lower limit value. The interrupt of overcurrent detection is output by these results.

It has 3 sets of upper limit value and lower limit value for each channel. And the interrupts can be output from each comparison results.

In addition, the overcurrent detection function also has overcurrent detection window notification function that notifies within range or out of range from the comparison result of the upper limit and the lower limit.

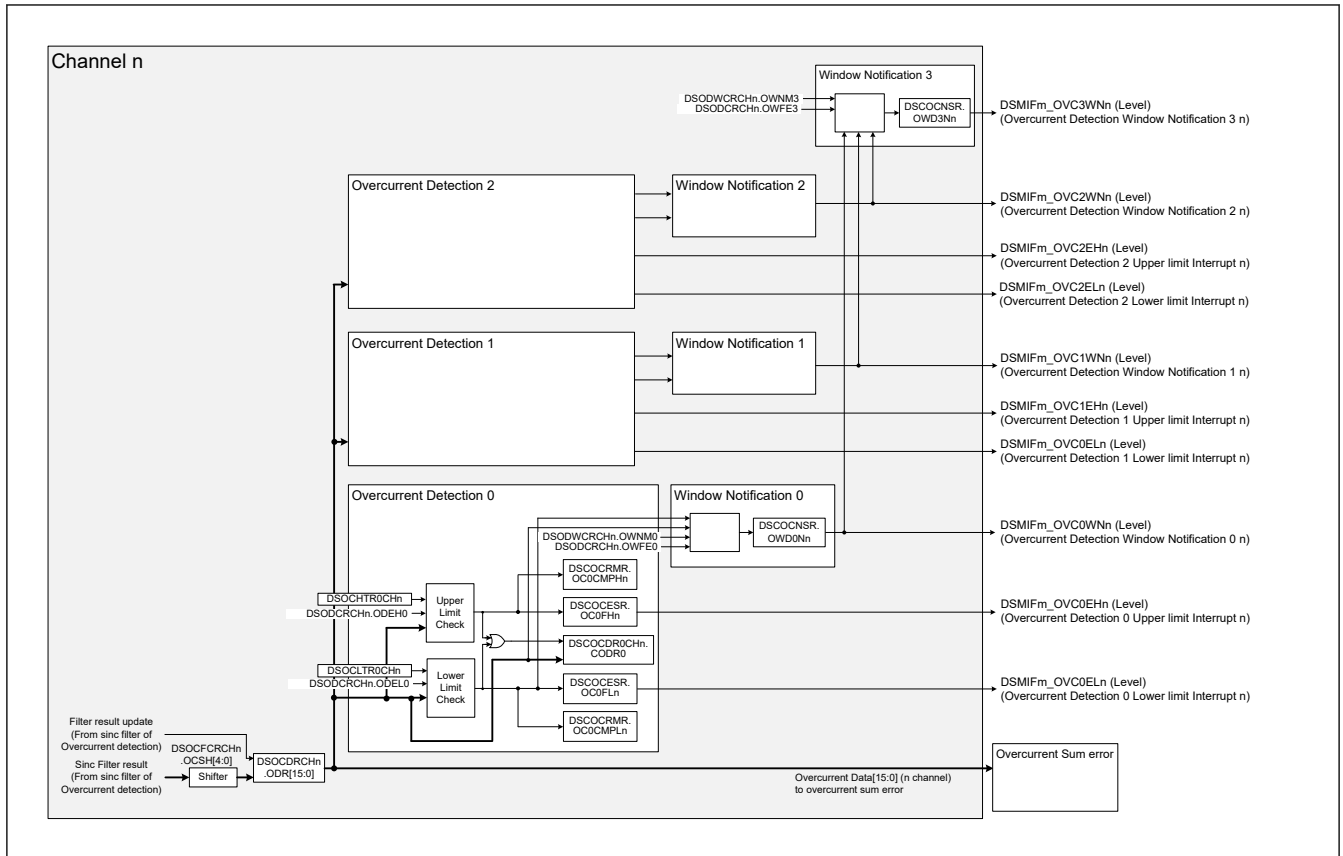


Figure 48.9 Configuration of overcurrent data registers

The measure values from current are stored as 16-bit values obtained by shifting the 25-bit data output from the sinc-filter in accordance with the setting of the DSOCFCRCHn.OC0SH[4:0] bits.

The format of stored data can be selected by the DSCMSR.DFS bit. Either left-justified or right-justified can be selected.

48.3.13.1 Overcurrent Detection 0 to 2

Overcurrent detection has three upper limit excess judgments and three lower limit judgments for each channel. Table 48.7 shows the list of overcurrent detection.

Table 48.7 List of overcurrent detection

Function	Upper Lower	Enable	Comparator table	Error flag	Comparator result	Interrupt	Error capture data
Overcurrent Detection 0	Upper	DSODCRCHn. ODEH0	DSOCHTR0CHn. OCMPTBH0	DSCOCESR. OC0FHn	DSCOCRMR. OC0CMPHn	OVC0EHn	DSCOCDR0CHn. CODR0
	Lower	DSODCRCHn. ODEL0	DSOCHTR0CHn. OCMPTBL0	DSCOCESR. OC0FLn	DSCOCRMR. OC0CMPLn	OVC0ELn	
Overcurrent Detection 1	Upper	DSODCRCHn. ODEH1	DSOCHTR1CHn. OCMPTBH1	DSCOCESR. OC1FHn	DSCOCRMR. OC1CMPHn	OVC1EHn	DSCOCDR1CHn. CODR1
	Lower	DSODCRCHn. ODEL1	DSOCHTR1CHn. OCMPTBL1	DSCOCESR. OC1FLn	DSCOCRMR. OC1CMPLn	OVC1ELn	
Overcurrent Detection 2	Upper	DSODCRCHn. ODEH2	DSOCHTR2CHn. OCMPTBH2	DSCOCESR. OC2FHn	DSCOCRMR. OC2CMPHn	OVC2EHn	DSCOCDR2CHn. CODR2
	Lower	DSODCRCHn. ODEL2	DSOCHTR2CHn. OCMPTBL2	DSCOCESR. OC2FLn	DSCOCRMR. OC2CMPLn	OVC2ELn	

The overcurrent detection function is explained using overcurrent detection 0 as an example.

When the overcurrent upper limit detection 0 enable (DSODCRCHn.ODEH0) is valid, the overcurrent upper limit detection can operate. And when the overcurrent lower limit detection 0 enable (DSODCRCHn.ODEL0) is valid, the overcurrent lower limit detection can operate.

The current data is stored in the DSODCRCHn as a 16-bit data shifted from 25-bit data output from sinc filter for overcurrent detection in accordance with the setting of the DSOCFCRCHn.OC SH[4:0] bits.

An overcurrent upper limit detection interrupt is requested when the value in the DSODCRCHn register is greater than the value of OCMPTBH0.

An overcurrent lower limit detection interrupt is requested when the value in the DSODCRCHn register is less than the value of OCMPTBL0.

Overcurrent detection has the following two statuses.

1. Error flags that are set when the upper limit is exceeded / less than the lower limit is detected and cleared with the clear bit.
2. Monitor bits that return the comparison result for each decimation cycle.

48.3.13.2 Overcurrent Detection Window Notification

Overcurrent detection window notification is event notification function to notify within a range or out of range by comparison with upper limit and lower limit of overcurrent detection. There are four kinds of notification for each channel. Table 48.8 shows a list of overcurrent detection notification.

Table 48.8 List of overcurrent detection notification

Function	Enable	Mode bit	Flag	Notification
Overcurrent detection Window notification 0	DSODCRCHn. OWFE0	DSODWCRCHn. OWNM0	DSCOCNSR. OWD0Nn	OVC0WNn
Overcurrent detection Window notification 1	DSODCRCHn. OWFE1	DSODWCRCHn. OWNM1	DSCOCNSR. OWD1Nn	OVC1WNn
Overcurrent detection Window notification 2	DSODCRCHn. OWFE2	DSODWCRCHn. OWNM2	DSCOCNSR. OWD2Nn	OVC2WNn
Overcurrent detection Window notification 3	DSODCRCHn. OWFE3	DSODWCRCHn. OWNM3	DSCOCNSR. OWD3Nn	OVC3WNn

Overcurrent detection window notification 0 to 2 will be generated by comparison with upper limit and lower limit for overcurrent detection 0 to 2. The notification can be set for within a range or out of range as Figure 48.10.

1. $OWNM_m = 0$: (overcurrent data channel $n < \text{lower limit } m$) or (overcurrent data channel $n > \text{upper limit } m$)
2. $OWNM_m = 1$: $\text{lower limit } m \leq \text{overcurrent data channel } n \leq \text{upper limit } m$

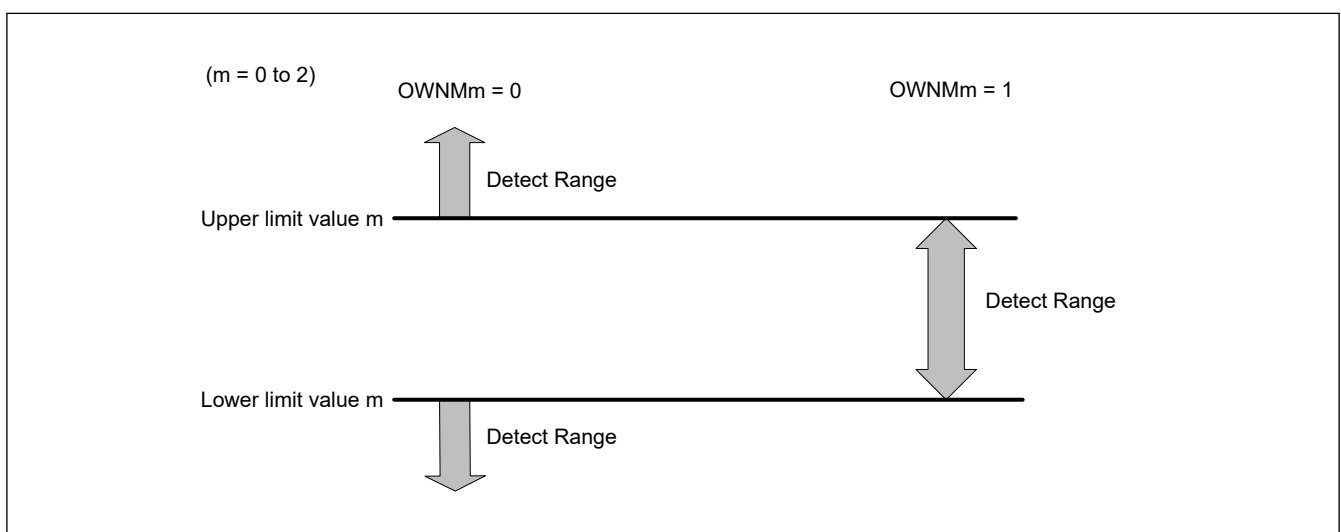


Figure 48.10 Detect range of overcurrent detection window notification

48.3.14 Overcurrent Sum Error Detection

When the overcurrent sum error upper limit detection enable (DSSECR.SEEH) is valid, the overcurrent sum error upper limit detection can operate. When the overcurrent sum error lower limit detection enable (DSSECR.SEEL) is valid, the

overcurrent sum error lower limit detection can operate. An overcurrent upper limit sum error detection interrupt is requested when the sum of the values for each mode (see Table 48.9) is higher than the value of SCMPTBH.

An overcurrent lower limit sum error detection interrupt is requested when the sum of the values for each mode is lower than the value of SCMPCTL.

Table 48.9 List of sum error mode detect mode and capture data

SEDM[2:0]	Sum error detect mode	CH0	CH1	CH2	DSSECDR0	DSSECDR1	DSSECDR2
000b	Three value sum error detect	✓	✓	✓	DSOCDRCH0	DSOCDRCH1	DSOCDRCH2
001b	Two value sum error detect	✓	✓	—	DSOCDRCH0	DSOCDRCH1	—*1
010b	One channel error detect	✓	—	—	DSOCDRCH0	—	—
011b		—	✓	—	DSOCDRCH1	—	—
100b		—	—	✓	DSOCDRCH2	—	—

Note 1. Do not store data, hold previous value.

48.3.15 Interrupt Function

DSMIF has the following interrupt function.

Table 48.10 DSMIF Interrupt List

No.	Category	Function	Interrupt name	Related bit name	Number / Unit	Pulse or Level
1	Data Update	Current data	Current Data Update	DSCSR.DUFn DSCSRCHn.DUF	3	Pulse
2			Current Data Update channel common	DSCSR.DUFn DSCSRCHn.DUF	1	Pulse
3		Capture data A	Capture data A Update	DSCSR.CAUFn DSCSRCHn.CAUF	3	Pulse
4			Capture data A Update channel common	DSCSR.CAUFn DSCSRCHn.CAUF	1	Pulse
5		Capture data B	Capture data B Update	DSCSR.CBUFn DSCSRCHn.CBUF	3	Pulse
6			Capture data B Update channel common	DSCSR.CBUFn DSCSRCHn.CBUF	1	Pulse
10	Error	Short-circuit	Short-circuit detection	DSCESR.SCFn DSCSRCHn.SCF	3	Level
11		Overcurrent detection	Overcurrent upper limit detection 0	DSCOCESR.OC0Fh DSCSRCHn.OC0FH	3	Level
12			Overcurrent lower limit detection 0	DSCOCESR.OC0FLn DSCSRCHn.OC0FL	3	Level
13			Overcurrent upper limit detection 1	DSCOCESR.OC1Fh DSCSRCHn.OC1FH	3	Level
14			Overcurrent lower limit detection 1	DSCOCESR.OC1FLn DSCSRCHn.OC1FL	3	Level
15			Overcurrent upper limit detection 2	DSCOCESR.OC2Fh DSCSRCHn.OC2FH	3	Level
16			Overcurrent lower limit detection 2	DSCOCESR.OC2FLn DSCSRCHn.OC2FL	3	Level
17		Overcurrent sum error	Overcurrent upper limit sum error detection	DSCESR.SUMERRH	1	Level
18			Overcurrent lower limit sum error detection	DSCESR.SUMERRL	1	Level

48.3.15.1 Data Update Interrupt

Each current data, capture current data A, and capture current data B have data update interrupts.

Table 48.11 shows summarizes the relationship between the data and flags related to each interrupt.

The operation of the current data update interrupt and the capture current data A / B update interrupt is explained in [section 48.3.15.1.1. Current Data Update Interrupt](#) and [section 48.3.15.1.2. Capture Current Data A/B Update Interrupt](#).

Table 48.11 List of Data update interrupt and related registers

Interrupt	Interrupt enable	Target data	Data update	
			Status flag	Flag Clear
Current Data Update	DSICRCHn.IUE	DSCDRCHn.ADDR	DSCSR.DUFn DSCSRCHn.DUF	DSCSCR.CLRDUFn DSCSRCHn.CLRDUF
Capture Current data A Update	DSICRCHn.IAUE	DSCCDRACHn.CDRA	DSCSR.CAUFn DSCSRCHn.CAUF	DSCSCR.CLRCAUF DSCSRCHn.CLRCAUF
Capture Current data B Update	DSICRCHn.IBUE	DSCCDRBCHn.CDRB	DSCSR.CBUFn DSCSRCHn.CBUF	DSCSCR.CLRCBUF DSCSRCHn.CLRCBUF

48.3.15.1.1 Current Data Update Interrupt

Each time current data register(DSCDRCHn) is updated, current data update interrupt is occurred.

This interrupt is a pulse interrupt. This interrupt can be enabled by DSICRCHn.IUE.

When this interrupt is occurred, Current data register is updated and current data valid flag (DSCSR.DUFn / DSCSRCHn.DUF) is set. Current data update flag can be cleared by the following condition (see [Table 48.12](#)).

Table 48.12 List of clear condition of Flag of Current Data

No.	Status Flag	Clear condition	Remarks
1	DSCSR.DUFn DSCSRCHn.DUF	Write 1 to DSCSCR.CLRDUFn or DSCSRCHn.CLRDUF	
2		Read DSCDRCHn	Even if DSCDRCHn is read, the value of DSCDRCHn is not cleared.

Figure 48.11 shows the operation of the current data update interrupt and the current data update flag.

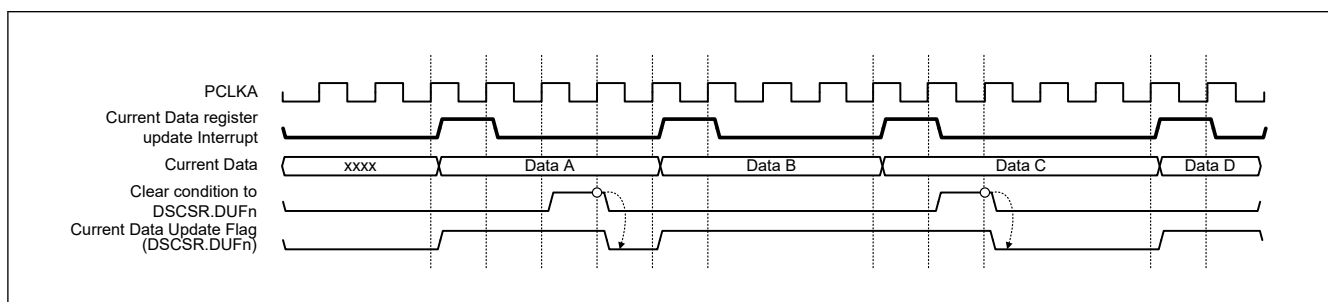


Figure 48.11 Operation example of interrupt and status register (Current Data register Update)

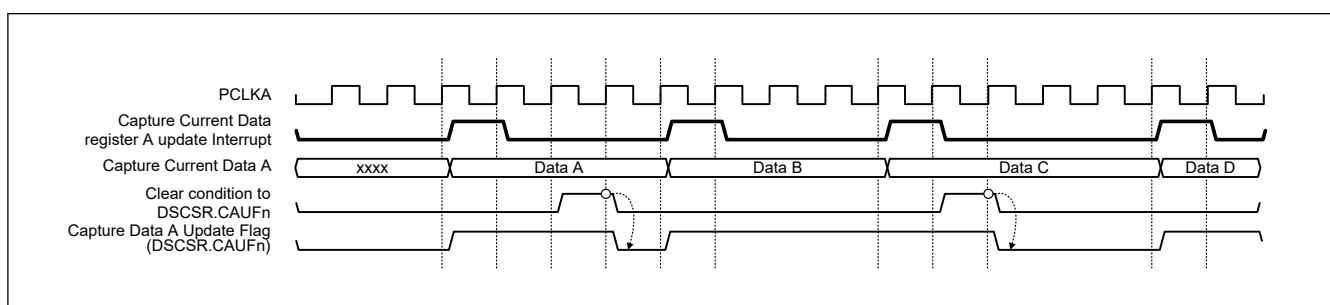
48.3.15.1.2 Capture Current Data A/B Update Interrupt

Each time capture current data register A/B(DSCCDRACHn.CDRA / DSCCDRBCHn.CDRB) is updated, capture current data A/B update interrupt is occurred. This interrupt is a pulse interrupt. This interrupt can be enabled by DSICRCHn.IAUE / DSICRCHn.IBUE. Capture data A/B update flag can be cleared by the following condition (see [Table 48.13](#)).

Table 48.13 List of clear condition of Flag of Capture Current Data A

No	Status Flag	Clear condition	Remarks
1	DSCSR.CAUFn DSCSRCHn.CAUF	Write 1 to DSCSCR.CLRCaufn or DSCSCRCHn.CLRCauf	
2		Read DSCCdrachn	Even if DSCCdrachn is read, the value of DSCCdrachn is not cleared.
3	DSCSR.CBUFn DSCSRCHn.CBUF	Write 1 to DSCSCR.CLRCBUFn or DSCSCRCHn.CLRCBUF	
4		Read DSCCdrbchn	Even if DSCCdrbchn is read, the value of DSCCdrbchn is not cleared.

Figure 48.12 shows the operation of the capture data A/B update interrupt and the capture current data update flag.

**Figure 48.12 Operation example of interrupt and status register (Capture Current Data register A Update Interrupt)**

48.3.15.1.3 Common Interrupt

The common interrupt is a function that outputs an interrupt that is common to one interrupt that is output for each channel. Common interrupts are provided for the following data update interrupts.

When using common interrupts, it is recommended to disable channel's data update interrupts.

Table 48.14 List of data update common interrupt

No.	Function	Interrupt name	Number / Unit	Pulse or Level	Interrupt enable
1	Current data	Current Data register Update Common Interrupt	1	Pulse	DSCICR.IUCE
2	Capture data A	Current Capture Data register A Update Common Interrupt	1	Pulse	DSCICR.IAUCE
3	Capture data B	Current Capture Data register B Update Common Interrupt	1	Pulse	DSCICR.IBUCE

Common interrupts have two modes, 3ch synchronization mode and 2ch synchronization mode.

As shown in Table 48.15, common interrupts can be output when synchronization mode is set and the all target channel's data is updated when.

Table 48.15 Mode List of common interrupt

CISM[2:0]	Function	Target channel			Target function		
		CH0	CH1	CH2	Current data	Capture data A	Capture data B
00B	Common interrupts can not be used.	—	—	—	—	—	—
01B	Three channel synchronous mode	✓	✓	✓	✓	✓	✓
10B	Two channel synchronous mode	✓	✓	—	✓	✓	✓
11B	Setting prohibited	—	—	—	—	—	—

The operation timing of the common interrupt in the three channel synchronization mode is explained using the current data update common interrupt as an example. When the current data is updated for each of the three channels, the current data common interrupt is output at the same timing as the latest channel current data update interrupt.

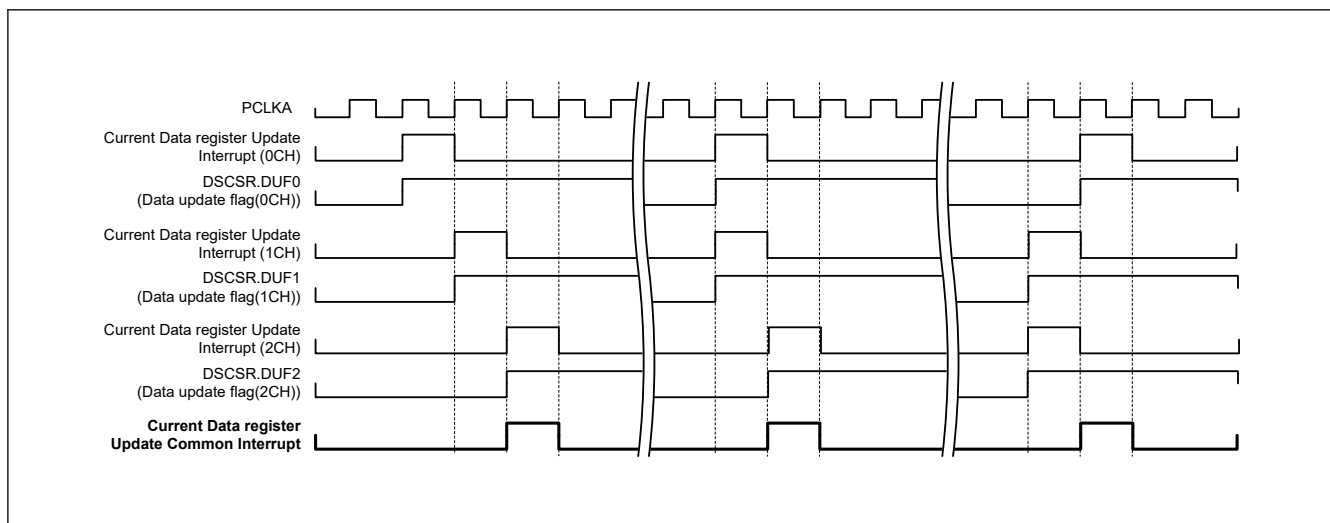


Figure 48.13 Common interrupt output timing (Three channel synchronous mode)

Similarly, in the two-channel synchronous mode, the current data common interrupt is output at the same timing as the latest channel current data update interrupt.

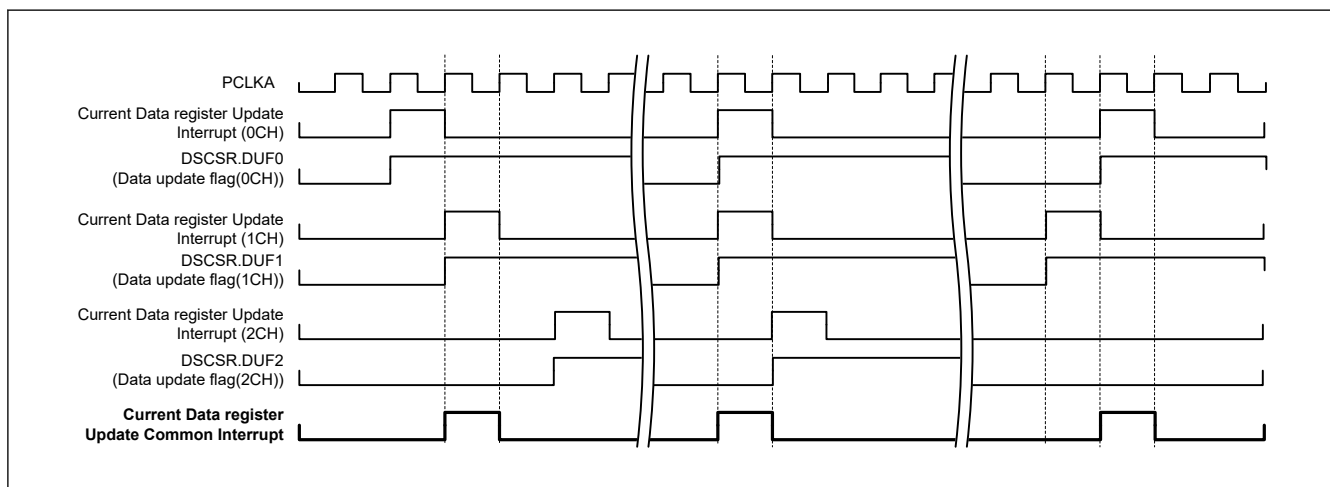


Figure 48.14 Common interrupt output timing (Two channel synchronous mode)

48.3.15.2 Error Interrupt

When error condition is detected, error interrupt is occurred. Error interrupt is a level interrupt.

When the corresponding error bit is cleared, the interrupt is set low level.

Figure 48.15 shows overcurrent upper limit error interrupt of error interrupt as an example.

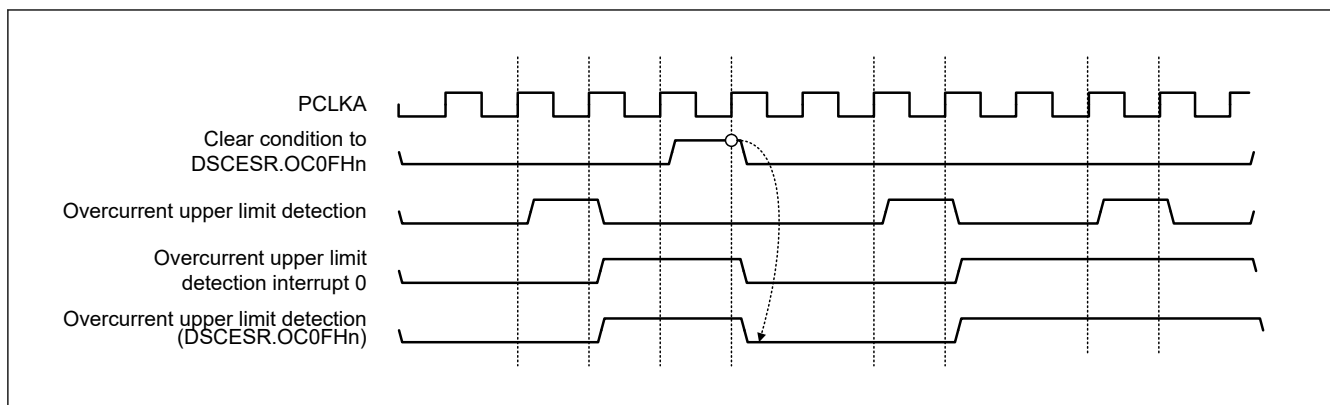


Figure 48.15 Operation of interrupt and status register (Error Interrupt)

48.3.16 Window Notification

Window notification is an event notification function that outputs for each channel.

Window notifications are provided for each channel, as shown below.

Table 48.16 List of O Window Notification

No.	Function	Notification name	Related bit name	Number / Unit	Pulse or Level	Notification enable
1	Overcurrent detection	Overcurrent Detection Window Notification 0	DSCOCNSR.OWD0N0 DSCSRCHn.OWD0N	3	Level	DSICRCHn.OWNE0
2		Overcurrent Detection Window Notification 1	DSCOCNSR.OWD0N1 DSCSRCHn.OWD1N	3	Level	DSICRCHn.OWNE1
3		Overcurrent Detection Window Notification 2	DSCOCNSR.OWD0N2 DSCSRCHn.OWD2N	3	Level	DSICRCHn.OWNE2
4		Overcurrent Detection Window Notification 3	DSCOCNSR.OWD0N3 DSCSRCHn.OWD3N	3	Level	DSICRCHn.OWNE3

48.3.16.1 Overcurrent Detection Window Notification

Overcurrent detection window notification is a level interrupt. When the corresponding status bit is cleared, the notification is set low level.

Figure 48.16 shows overcurrent detection window notification 0 as an example.

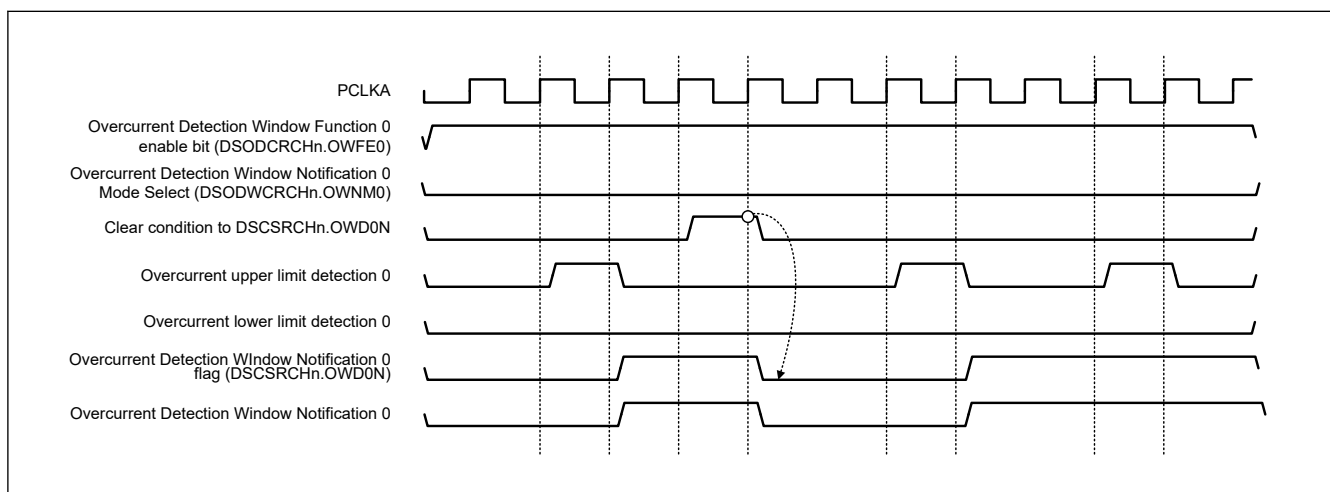


Figure 48.16 Operation of Overcurrent Detection Window Notification

48.3.17 Settling Time

48.3.17.1 Settling Time of Channel Activation

Settling time of channel activation is the time from when the channel is activated until the filter result is stable and output. The settling time is calculated as follows.

Table 48.17 Settling time of channel activation

Core clock select (DSCCSR.CLKSEL)	Sinc order	Processing time of Settling time of channel activation
00b (PCLKA and CCLK are synchronous clocks)	Sinc ¹	(DSMCLK period * Decimation ratio) * sinc order + DSMCLK period * 1 cycle + CCLK period * 12 cycle + PCLKA period * 6 cycle
	Sinc ³ Sinc ²	(DSMCLK period * Decimation ratio) * (sinc order + 1) + DSMCLK period * 1 cycle + CCLK period * 12 cycle + PCLKA period * 6 cycle (*1)
01b or 10b (PCLKA and CCLK are asynchronous clocks)	Sinc ¹	(DSMCLK period * Decimation ratio) * sinc order + DSMCLK period * 1cycle + CCLK period * 21 cycle + PCLKA period * 11 cycle
	Sinc ³ Sinc ²	(DSMCLK period * Decimation ratio) * (sinc order + 1) + DSMCLK period * 1cycle + CCLK period * 21cycle + PCLKA period * 11cycle

48.3.17.2 Settling Time after PWM Synchronization

Settling time after PWM synchronization is the time from the input of the decimation dividing counter initialization trigger to the stable output of the filter result.

Table 48.18 Settling time after PWM synchronization

Core clock select (DSCCSR.CLKSEL)	Processing time of Settling time of PWM synchronization
00b (PCLKA and CCLK are synchronous clocks)	(DSMCLK period * Decimation ratio) * sinc order + CCLK period * 9cycle + PCLKA period * 4cycle
01b or 10b (PCLKA and CCLK are asynchronous clocks)	(DSMCLK period * Decimation ratio) * sinc order + CCLK period * 15cycle + PCLKA period * 7cycle

48.4 Usage Note

48.4.1 Clock Frequency Selection

When the frequency of the core clock and bus clock is changed, change it after stopping the DSMIF according to Stop flow. It is prohibited to change the clock frequency while DSMIF is operating.

48.4.2 Software Standby Mode

Before transition to software standby mode, stop operation according to Stop Flow.

After return from software standby mode, start operation according to Start Flow. Register setting before transition is retained. Therefore, you can omit the setting that is not needed to be changed.